

Let, the things be adjusted such that x_f' is identically equal to the externally applied input signal x_i . Now if the external source x_i is removed and terminal 2 is connected to terminal 1, as shown in Fig. 8.2, the amplifier will continue to give the same output as before. We say that the amplifier is producing an output on its own or it is oscillating.

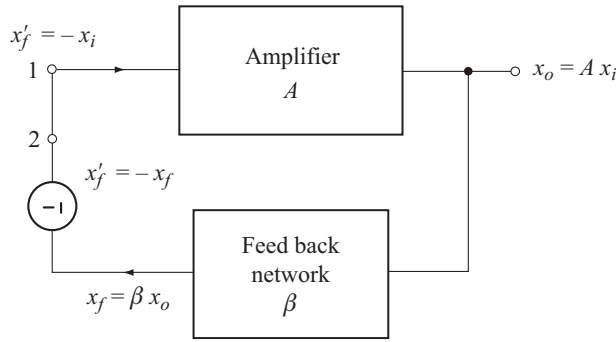


Fig. 8.2 Block diagram of oscillator

$x_f' = x_i$ means that, the instantaneous values of x_f' and x_i are exactly equal at all times. Since x_f' and x_i are in phase, the feedback is positive.

The condition $x_f' = x_i$ is satisfied only by the sine wave since, it preserves its shape when passed through any linear network. Thus the set up shown in Fig. 8.2 produces sinusoidal oscillations.

Substituting $x_f' = x_i$ in Equation (8.1) we have

$$\text{Loop gain} = -A\beta = 1 \quad (8.2)$$

$$-A\beta = 1 + j0 \quad [\text{Rectangular form}]$$

$$\text{or} \quad -A\beta = 1 \angle 0^\circ \text{ or } 360^\circ \quad [\text{Polar form}] \quad (8.3)$$

$$\text{In general} \quad -A\beta = |-A\beta| \angle \theta \quad (8.4)$$

Comparing Equations (8.3) and (8.4) we get

$$|-A\beta| = |A\beta| = 1 \quad (8.5)$$

$$\text{and} \quad \theta = \angle -A\beta = 0^\circ \text{ or } 360^\circ \quad (8.6)$$

The conditions given in Equations (8.5) and (8.6) are called Barkhausen Criteria for sustained oscillations. These conditions are stated as follows.

- **The frequency at which a sinusoidal oscillator will operate is the frequency for which the phase shift θ of the loop gain is 0° or 360° .**
- **Oscillations will be sustained at the oscillator frequency if the magnitude of loop gain is equal to unity.**

◆ 8.2 PRACTICAL ASPECTS

- In practical oscillators, initially the magnitude of loop gain is set to a value slightly greater than unity. As a result the amplitude of oscillations begins to grow. When the amplitude reaches the desired level, the automatic gain control mechanism reduces the gain so as to make the magnitude of loop gain become equal to unity.
- The frequency of oscillation is decided by the elements of the feedback network.
- The function of inverting network is performed by the feedback network itself. i.e., the feedback network introduces a phase shift of 180° . The amplifier produces a phase shift of 180° . As a result the total phase shift becomes 360° satisfying the angle criterion of Barkhausen.

◆ 8.3 CLASSIFICATION OF OSCILLATORS BASED ON THE ELEMENTS USED IN THE FEEDBACK NETWORK

Based on the elements used in the feedback network, oscillators can be classified as follows.

- RC oscillators
 - LC oscillators
 - Crystal oscillators.
- In RC oscillators, the feedback network uses RC components to generate oscillations. RC oscillators are used to generate oscillations in the audio frequency range. [20 Hz – 20 kHz].
 - In LC oscillators, the feedback network employs LC components to generate oscillations. LC oscillators are used to generate oscillations in the radio frequency range [100 kHz – 100 MHz].
 - In crystal oscillators, the feedback network uses piezo electric crystal to generate oscillations. Crystal oscillators are used to generate oscillations in the frequency range [10 kHz to 10 MHz].

◆ 8.4 RC-PHASE SHIFT OSCILLATOR

RC-phase shift oscillator is a low frequency oscillator. It is used to generate oscillations in the audio frequency range. The feedback network consists of three identical RC-sections as shown in Fig. 8.3.

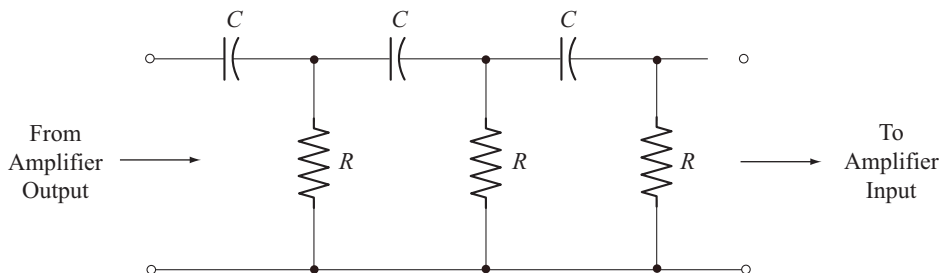


Fig. 8.3 Feedback network in RC-phase shift oscillator

Figure 8.4 (a) shows a single RC-section and its phasor diagram is shown in Fig. 8.4 (b).

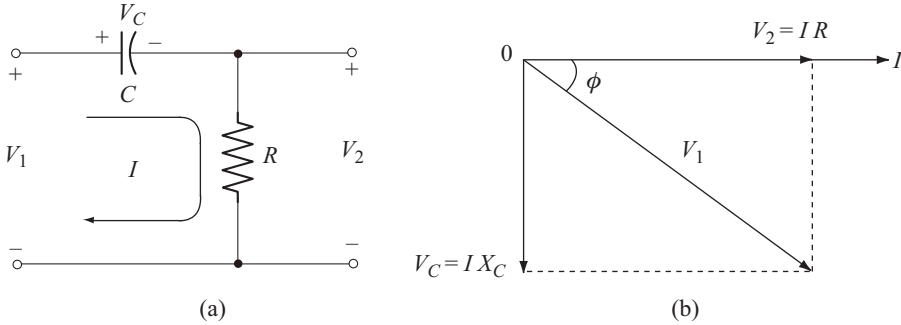


Fig. 8.4 (a) Single RC-section
(b) Phasor diagram

The phasor diagram is drawn taking the current I as the reference vector and using the fact that, in a resistor, the voltage and current are in phase and in a capacitor, the current leads the voltage by an angle 90° . Note that the voltage V_2 leads V_1 by an angle ϕ .

From the phasor diagram

$$\tan \phi = \frac{V_C}{V_2} = \frac{IX_C}{IR} = \frac{X_C}{R}$$

$$\text{Using } X_C = \frac{1}{\omega C} = \frac{1}{2\pi f_c}$$

$$\tan \phi = \frac{1}{2\pi f R C} \quad (8.7)$$

The values of R and C are selected so as to give a phase shift of 60° at the desired frequency of oscillations. Since all the three RC-sections are identical, the total phase shift introduced by the feedback network is 180° .

◆ 8.5 TRANSISTOR RC-PHASE SHIFT OSCILLATOR

Figure 8.5 shows the circuit of transistor RC-phase shift oscillator. It consists of a single stage RC-coupled CE amplifier and a feedback network comprising of three identical RC-sections.

The input to the feedback network is the amplifier output voltage V_o and the output of the feedback network is the current I_f' . Note that the circuit employs voltage sampling and current mixing (shunt mixing). Hence the topology is voltage shunt feedback.

If the output voltage of the feedback network were directly connected to the amplifier input, the relatively low input resistance (h_{ie}) of the amplifier appreciably loads down the feedback network. Hence voltage shunt feedback is more suitable.

The resistor R' of the last RC-section is returned to the ground via the input resistance of amplifier stage. Thus the total resistance of the last RC-section is $R' + h_{ie}$. For the three RC-sections to be identical it is essential that

$$R = R' + h_{ie} \quad (8.8)$$

$$\text{or} \quad R' = R - h_{ie} \quad (8.9)$$

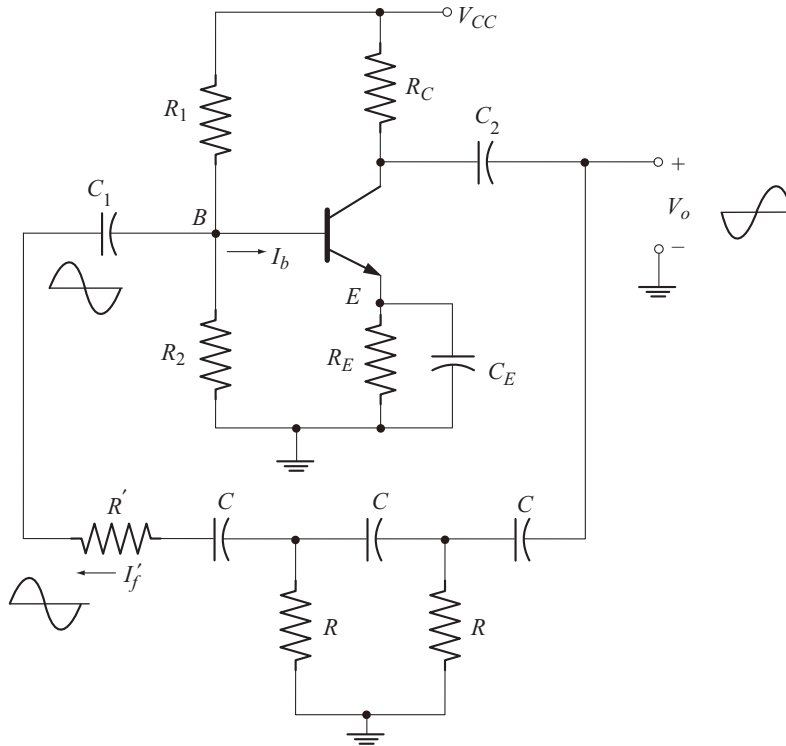


Fig. 8.5 Transistor RC-phase shift oscillator

The values of R and C are selected such that each RC-section gives a phase shift of 60° at the oscillator frequency. The total phase shift from the three RC-sections will be 180° . The CE stage introduces a phase shift of 180° . As a result the total phase shift around the loop is 360° , thus satisfying the angle criteria.

The frequency of oscillations is given by

$$f = \frac{1}{2\pi RC\sqrt{6+4k}} \quad (8.10)$$

$$\text{where} \quad k = \frac{R_C}{R} \quad (8.11)$$

$$\text{Loop gain} = \frac{x'_f}{x_i} = \frac{I'_f}{I_b} \quad (8.12)$$

For the magnitude of loop gain to be greater than unity, the short circuit current gain h_{fe} of the transistor must satisfy the condition

$$h_{fe} > 4k + 23 + \frac{29}{k} \quad (8.13)$$

The minimum value of h_{fe} can be obtained by differentiating h_{fe} with respect to k and equating the result to zero. This yields $h_{fe(\min)} = 44.5$ corresponding to $k = 2.7$.

A transistor with a small-signal common-emitter short circuit current gain less than 44.5 cannot be used in the phase-shift oscillator.

◆ 8.6 MERITS AND DEMERITS OF RC-PHASE SHIFT OSCILLATOR

The merits of RC-phase shift oscillator are:

- The circuit is simple and easy to design.
- The circuit is less bulky, light weight and less expensive due to the absence of inductor.
- It can be used to generate oscillations in the audio frequency range.
- It generates sinusoidal oscillations.

The demerits are:

- RC network has relatively low quality factor. This results in frequency drift when the circuit parameter values vary due to ageing, temperature, etc.. Hence this circuit has poor frequency stability.
- To generate high frequencies, the required feedback network capacitance becomes too small which will be comparable with the stray capacitances. Hence it has no marked advantage over LC oscillators at high frequencies.
- Variable frequency operation is obtained by changing the three capacitors simultaneously. Such a variation should keep the impedance of the phase shifting network constant and keeps the magnitude of β and $A\beta$ constant in order to have constant amplitude of oscillations at all frequencies. But it is very difficult to meet the above requirement over a wide frequency range.

Example 8.1

In an RC-phase shift oscillator $R = 1 \text{ k}\Omega$, $R_C = 1 \text{ k}\Omega$ and $C = 0.1 \text{ }\mu\text{F}$. Calculate the frequency of oscillations.

Solution

$$f = \frac{1}{2\pi RC\sqrt{6+4k}}$$

$$k = \frac{R_C}{R} = \frac{1 \text{ k}\Omega}{1 \text{ k}\Omega} = 1$$

$$f = \frac{1}{2\pi (1 \text{ k}\Omega)(0.1 \text{ }\mu\text{F})\sqrt{6+4}} = 503.29 \text{ Hz}$$

Example 8.2

In an RC-phase shift oscillator $R_C = 3.3 \text{ k}\Omega$ and $R = 1.5 \text{ k}\Omega$. Calculate the required value of C for a frequency of 2 kHz.

Solution

$$\begin{aligned}
 f &= \frac{1}{2\pi RC\sqrt{6+4k}} \\
 \Rightarrow C &= \frac{1}{2\pi f R\sqrt{6+4k}} \\
 k &= \frac{R_C}{R} = \frac{3.3 \text{ k}\Omega}{1.5 \text{ k}\Omega} = 2.2 \\
 C &= \frac{1}{2\pi(2000)(1.5 \text{ k}\Omega)\sqrt{6+4(2.2)}} = 13.79 \text{ nF}
 \end{aligned}$$

Example 8.3

An RC-phase shift oscillator uses a transistor with $h_{fe} = 100$. If $R_C = 10 \text{ k}\Omega$ and $R = 2 \text{ k}\Omega$. Will this circuit oscillate?

Solution

Condition for sustained oscillations is

$$\begin{aligned}
 h_{fe} &> 4k + 23 + \frac{29}{k} \\
 k &= \frac{R_C}{R} = \frac{10 \text{ k}\Omega}{2 \text{ k}\Omega} = 5 \\
 h_{fe} &> (4)(5) + 23 + \frac{29}{5} \\
 h_{fe} &> 48.8
 \end{aligned}$$

Since the h_{fe} of the transistor is 100 which is greater than 48.8, the circuit oscillates.

Example 8.4

In an RC-phase shift oscillator $R_C = 5 \text{ k}\Omega$ and $R = 3.3 \text{ k}\Omega$. Find the range of values of C if it is required to vary the frequency from 100 Hz to 20 kHz.

Solution

$$\begin{aligned}
 f &= \frac{1}{2\pi RC\sqrt{6+4k}} \\
 \Rightarrow C &= \frac{1}{2\pi f R\sqrt{6+4k}}
 \end{aligned}$$

$$k = \frac{R_C}{R} = \frac{5 \text{ k}\Omega}{3.3 \text{ k}\Omega} = 1.515$$

$$\sqrt{6+4k} = \sqrt{6+(4)(1.515)} = 3.472$$

For $f = 100 \text{ Hz}$

$$C = \frac{1}{2\pi(100)(3.3 \text{ k}\Omega)(3.472)} = 138.9 \text{ nF}$$

For $f = 20 \text{ kHz}$

$$C = \frac{1}{2\pi(20,000)(3.3 \text{ k}\Omega)(3.472)} = 0.6945 \text{ nF}$$

Range of C value: 0.6945 nF to 138.9 nF

Example 8.5

Find the values of R_C , R , R' and C for an RC-phase shift oscillator for a frequency of oscillation of 1000 Hz. A transistor with $h_{fe} = 200$ and $h_{ie} = 2 \text{ k}\Omega$ is available.

Solution

Selection of R and R'

$$R = R' + h_{ie}$$

$$R' = R - h_{ie}$$

For R' to be positive,

$$R > h_{ie}$$

i.e., $R > 2 \text{ k}\Omega$

Let $R = 2.7 \text{ k}\Omega$

Now $R' = 2.7 \text{ k}\Omega - 2 \text{ k}\Omega$

$$R' = 700\Omega$$

Selection of R_C

The condition to get sustained oscillations is

$$h_{fe} > 4k + 23 + \frac{29}{k}$$

$$200 > 4k + 23 + \frac{29}{k}$$

We are free to choose the value of k to meet the above requirement.

With $k = 1$, we have

$$200 > 4 + 23 + 29$$

$$200 > 56$$

$k=1$ satisfies the condition for sustained oscillations.

$$k = \frac{R_C}{R}$$

$$R_C = k R = 2.7 \text{ k}\Omega$$

$$C = \frac{1}{2\pi R f \sqrt{6 + 4k}} = 18.6 \text{ nF}$$

Note: R_1 , R_2 and R_E can be calculated using the biasing requirement.

Example 8.6

For the transistor RC-phase shift oscillator, derive:

- the expression for frequency of oscillations.
- the condition required for sustained oscillations.

Solution

(a) Expression for frequency of oscillations

The output equivalent circuit of the phase-shift oscillator of Fig. 8.5 is shown in Fig. A. Note that the resistor R' is returned to ground via the input resistance h_{ie} of the amplifier stage. $\frac{1}{h_{oe}}$ is treated as an open circuit assuming $h_{oe} \approx 0$.

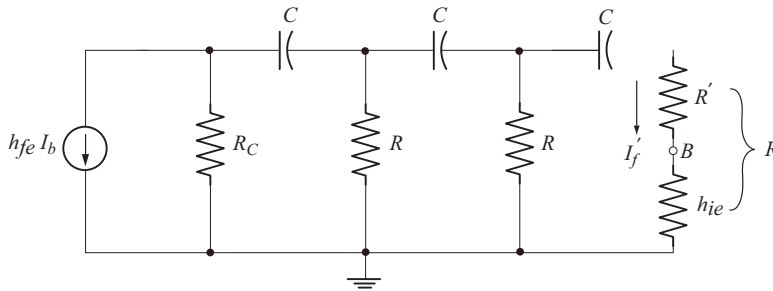


Fig. A

Let us convert the current source into its equivalent voltage source as shown in Fig. B.

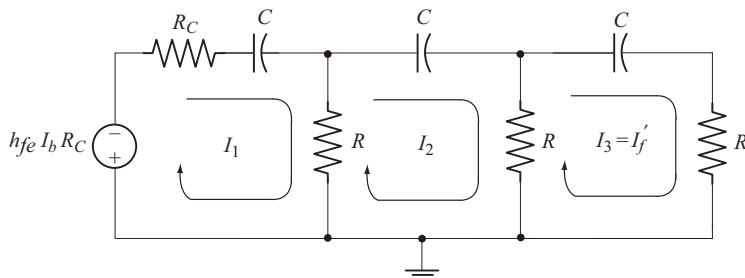


Fig. B

Loop 1

$$-h_{fe} I_b R_C = I_1 [R_C + R - j X_C] - I_2 R \quad (\text{A})$$

where $X_C = \frac{1}{\omega C}$

Dividing through out by R and taking

$$k = \frac{R_C}{R} \quad \text{and} \quad \alpha = \frac{X_C}{R} \quad \text{we have}$$

$$-h_{fe} I_b k = I_1 [k + 1 - j \alpha] - I_2 \quad (\text{B})$$

Loop 2

$$0 = -I_1 R + I_2 [2R - j X_C] - I_3 R$$

or $0 = -I_1 + I_2 [2 - j \alpha] - I_3 \quad (\text{C})$

Loop 3

$$0 = -I_2 R + I_3 [2R - j X_C]$$

or $0 = -I_2 + I_3 [2 - j \alpha]$

$$\Rightarrow I_2 = I_3 [2 - j \alpha] \quad (\text{D})$$

From Equation (C)

$$I_1 = I_2 [2 - j \alpha] - I_3$$

Substituting for I_2 from Equation (D) we have

$$I_1 = I_3 [2 - j \alpha]^2 - I_3$$

$$I_1 = I_3 [(2 - j \alpha)^2 - 1]$$

$$I_1 = I_3 [3 - \alpha^2 - j 4 \alpha] \quad (\text{E})$$

Substituting Equations (D) and (E) in Equation (B) we have

$$-h_{fe} I_b k = I_3 [3 - \alpha^2 - j 4 \alpha] [k + 1 - j \alpha] - I_3 [2 - j \alpha]$$

$$-h_{fe} I_b k = I_3 \{ [1 + 3k - (5 + k) \alpha^2] - j [(6 + 4k) \alpha - \alpha^3] \} \quad (\text{F})$$

$$\text{Loop gain} = \frac{I_f'}{I_b} = \frac{I_3}{I_b}$$

From Equation (F) $\frac{I_3}{I_b} = \frac{-h_{fe} k}{[1 + 3k - (5 + k) \alpha^2] - j [(6 + 4k) \alpha - \alpha^3]} \quad (\text{G})$

For the loop gain to be real

$$(6 + 4k) \alpha - \alpha^3 = 0 \quad (\text{H})$$

$$\alpha^2 = (6 + 4k) \quad (\text{I})$$

$$\alpha = \sqrt{6+4k} \quad (J)$$

But
$$\alpha = \frac{X_C}{R} = \frac{1}{\omega C R} = \frac{1}{2 \pi f R C}$$

Using this relation in Equation (J) we have

$$\frac{1}{2 \pi f R C} = \sqrt{6+4k}$$

$$f = \frac{1}{2 \pi R C \sqrt{6+4k}} \quad (K)$$

(b) Condition for sustained oscillations

Substituting Equation (H) in Equation (G) we get

or
$$\frac{I_3}{I_b} = \frac{-h_{fe} k}{1+3k-[5+k]\alpha^2} \quad (L)$$

$$Dr = 1+3k-[5+k]\alpha^2$$

Using $\alpha^2 = 6+4k$ we have

$$Dr = 1+3k-[5+k][6+4k]$$

$$Dr = -[4k^2+23k+29]$$

Now
$$\frac{|I_3|}{|I_b|} = \frac{h_{fe} k}{4k^2+23k+29}$$

$$= \frac{h_{fe}}{4k+23+\frac{29}{k}} \quad (M)$$

To get sustained oscillations the magnitude of loop gain should be greater than unity.

i.e.,
$$\frac{|I_3|}{|I_b|} > 1$$

$$\frac{h_{fe}}{4k+23+\frac{29}{k}} > 1$$

or
$$h_{fe} > 4k+23+\frac{29}{k} \quad (N)$$

Example 8.7

Show that for oscillations to start in an RC-phase shift oscillator, the minimum h_{fe} value of the transistor should be 44.5.

Solution

The condition required for sustained oscillations is

$$h_{fe} > 4k + 23 + \frac{29}{k} \quad (A)$$

h_{fe} is a function of k . Taking the equality condition in Equation (A), differentiating h_{fe} with respect to k and equating the result to zero we have

$$\begin{aligned} \frac{dh_{fe}}{dk} &= 4 - \frac{29}{k^2} = 0 \\ k^2 &= \frac{29}{4} \quad \text{or} \quad k = 2.7 \end{aligned}$$

Using this value in Equation (A) we have

$$\begin{aligned} h_{fe} &> 4(2.7) + 23 + \frac{29}{2.7} \\ h_{fe} &> 44.5 \\ \Rightarrow h_{fe(\min)} &= 44.5 \end{aligned}$$

◆ 8.7 WEIN BRIDGE OSCILLATOR

Wein bridge oscillator is an RC oscillator and is used to generate sinusoidal oscillations in the audio frequency range. Figure 8.6 shows the circuit of wein bridge oscillator.

It consists of an Op-Amp non inverting amplifier in the forward path and a lead-lag network in the feedback path. Series $R_1 C_1$ network is a lead network and the parallel $R_2 C_2$ network is the lag network. The name wein bridge is due to the fact that, the feedback resistors R_3 and R_4 of Op-Amp amplifier and the Lead-Lag network forms a bridge as shown in Fig. 8.7.

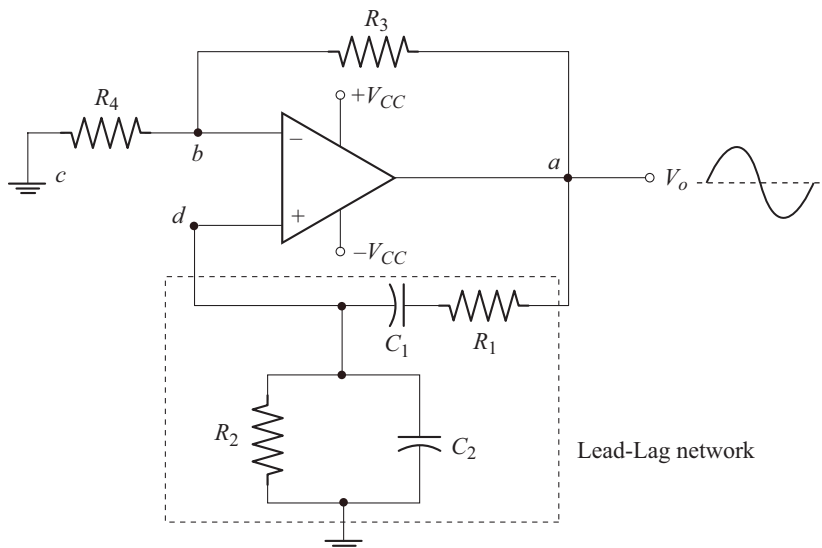


Fig. 8.6 Wein bridge oscillator using Op-Amp amplifier

At the oscillator frequency, the lead-lag network is designed to introduce zero degree phase shift. The Op-Amp non inverting amplifier introduces zero degree phase shift. Hence the total phase shift around the loop is zero.

The expression for the frequency of oscillation is obtained from the balancing condition of the bridge.

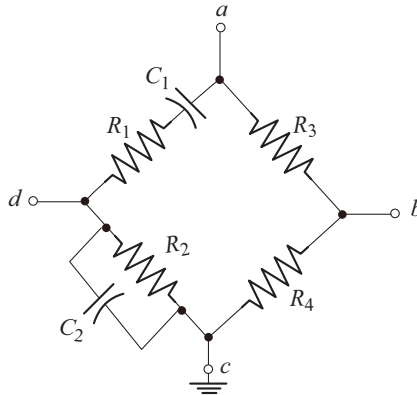


Fig. 8.7 Bridge circuit

The balancing condition is given by

$$\frac{R_3}{R_4} = \frac{R_1}{R_2} + \frac{C_2}{C_1}$$

and the frequency of oscillation is given by

$$f = \frac{1}{2\pi\sqrt{R_1 R_2 C_1 C_2}} \quad (8.14)$$

If $R_1 = R_2 = R$ and $C_1 = C_2 = C$, then

$$f = \frac{1}{2\pi RC} \quad (8.15)$$

In order to obtain sustained oscillations the gain of the amplifier should be at least equal to 3

$$\therefore 1 + \frac{R_3}{R_4} \geq 3$$

$$\text{or} \quad \frac{R_3}{R_4} \geq 2 \quad (8.16)$$

Example 8.8

The following component values are given for the wein bridge oscillator of the circuit of Fig. 8.6.

$$\begin{aligned} R_1 = R_2 &= 33 \text{ k}\Omega & C_1 = C_2 &= 0.001 \text{ }\mu\text{F} \\ R_3 &= 47 \text{ k}\Omega & R_4 &= 15 \text{ k}\Omega \end{aligned}$$

- (a) Will this circuit oscillate?
 (b) Calculate the resonant frequency.
 (c) Suggest the RC elements to increase the frequency by two fold.

Solution

$$(a) \quad \frac{R_3}{R_4} = \frac{47 \text{ k}\Omega}{15 \text{ k}\Omega} = 3.13 > 2$$

The circuit oscillates since $\frac{R_3}{R_4} > 2$.

$$(b) \quad R_1 = R_2 = R = 33 \text{ k}\Omega \quad C_1 = C_2 = C = 0.001 \text{ }\mu\text{F}$$

$$f = \frac{1}{2\pi RC}$$

$$= \frac{1}{2\pi (33 \text{ k}\Omega) (0.001 \text{ }\mu\text{F})} = 4.82 \text{ kHz}$$

$$f = 2 (4.82 \text{ kHz}) = 9.64 \text{ kHz}$$

$$f = \frac{1}{2\pi RC} \Rightarrow RC = \frac{1}{2\pi f}$$

Let us not change C

$$\therefore C = 0.001 \text{ }\mu\text{F}$$

$$\text{Now } R = \frac{1}{2\pi f C} = \frac{1}{2\pi (9.64 \text{ kHz}) (0.001 \text{ }\mu\text{F})} = 16.5 \text{ k}\Omega$$

Example 8.9

Design the component values of wein bridge oscillator of Fig. 8.6 for a frequency of oscillations of 4 kHz.

Solution

$$\text{Let } R_1 = R_2 = R \text{ and } C_1 = C_2 = C$$

$$f = \frac{1}{2\pi RC}$$

$$\text{Select } C = 0.01 \text{ }\mu\text{F}$$

$$R = \frac{1}{2\pi f C} = \frac{1}{2\pi (4 \text{ kHz}) (0.01 \text{ }\mu\text{F})} = 3.97 \text{ k}\Omega$$

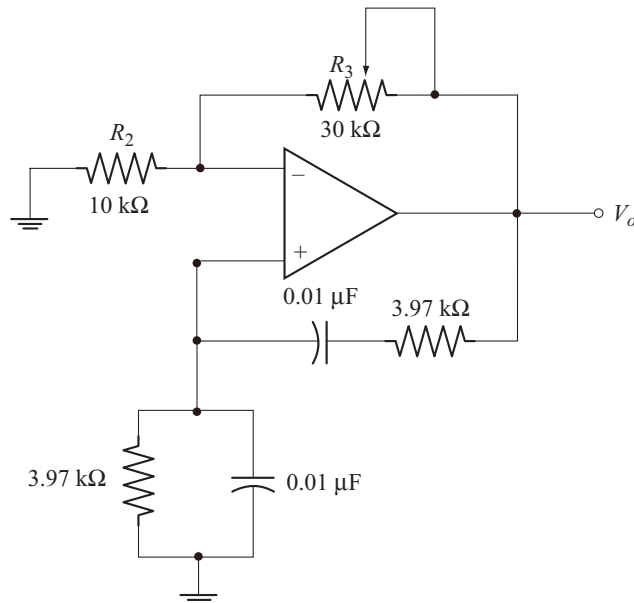
For sustained oscillations

$$\frac{R_3}{R_4} \geq 2$$

$$\begin{aligned}
 R_3 &\geq 2R_4 \\
 \text{Let } R_4 &= 10 \text{ k}\Omega \\
 \text{then } R_3 &\geq 20 \text{ k}\Omega \\
 \text{Select } R_3 &= 30 \text{ k}\Omega
 \end{aligned}$$

A 30 k Ω potentiometer can be used for R_3 . R_3 is adjusted until undistorted sustained oscillations are obtained.

The circuit diagram is shown below.



Example 8.10

For the wein bridge oscillator of Fig. 8.6 derive the expressions for

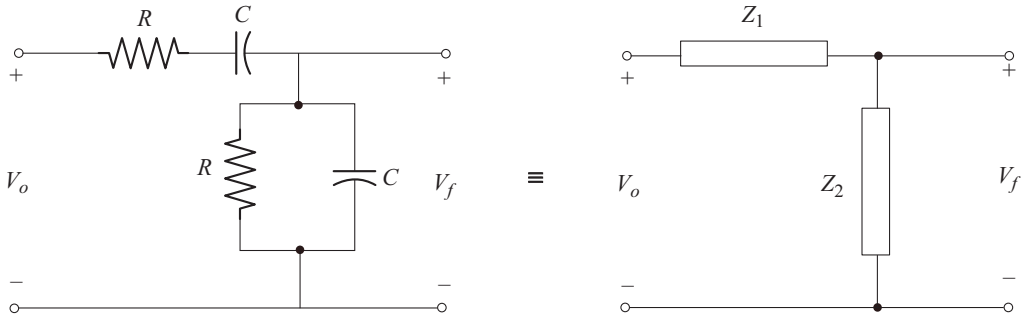
- Frequency of oscillations
- Condition for sustained oscillations

Solution

(a) Expression for Frequency of Oscillation

The feedback network of wein bridge oscillator is shown below. We have taken $R_1 = R_2 = R$ and $C_1 = C_2 = C$.

$$\begin{aligned}
 \text{Let } Z_1 &= R + \frac{1}{j\omega C} \\
 Z_1 &= \frac{1 + j\omega RC}{j\omega C} \quad (A)
 \end{aligned}$$



and

$$Z_2 = R \parallel \frac{1}{j\omega C}$$

$$= \frac{(R)\left(\frac{1}{j\omega C}\right)}{R + \frac{1}{j\omega C}}$$

$$Z_2 = \frac{R}{1 + j\omega RC} \quad (B)$$

Using voltage division rule we have

$$V_f = \frac{V_o Z_2}{Z_1 + Z_2}$$

Feedback factor is

$$\beta = \frac{V_f}{V_o} = \frac{Z_2}{Z_1 + Z_2}$$

$$\beta = \frac{\left[\frac{R}{1 + j\omega RC}\right]}{\left[\frac{1 + j\omega RC}{j\omega C}\right] + \left[\frac{R}{1 + j\omega RC}\right]}$$

$$= \frac{j\omega RC}{[1 + j\omega RC]^2 + j\omega RC}$$

$$= \frac{j\omega RC}{[1 - \omega^2 R^2 C^2] + j3\omega RC}$$

$$= \frac{j\omega RC}{j\{3\omega RC - j[1 - \omega^2 R^2 C^2]\}}$$

$$\beta = \frac{V_f}{V_o} = \frac{\omega RC}{3\omega RC + j[\omega^2 R^2 C^2 - 1]} \quad (C)$$

For zero phase shift between V_f and V_o , the required condition is

$$\omega^2 R^2 C^2 - 1 = 0 \quad (D)$$

$$\text{or} \quad \omega^2 R^2 C^2 = 1$$

$$\Rightarrow \quad \omega RC = 1$$

$$\omega = \frac{1}{RC}$$

$$\text{or} \quad f = \frac{1}{2\pi RC} \quad (E)$$

(b) Condition for Sustained Oscillations

Substituting equation (D) in equation (C) we get

$$\beta = \frac{1}{3} \quad (F)$$

For the oscillations to start the magnitude of loop gain must be at least equal to unity.

$$\text{i.e.,} \quad |A\beta| \geq 1$$

$$|A| \geq \frac{1}{\beta}$$

$$\text{or} \quad |A| \geq 3 \quad (G)$$

But for the Op-Amp non-inverting amplifier

$$|A| = A = 1 + \frac{R_3}{R_4}$$

Using this relation in equation (G) we have

$$1 + \frac{R_3}{R_4} \geq 3$$

$$\text{or} \quad \frac{R_3}{R_4} \geq 2 \quad (H)$$

Example 8.11

List the similarities and differences between RC-phase shift and wein bridge oscillators.

Solution

The similarities and differences between the RC-phase shift and wein bridge oscillators are given in the following table.

S. No.	RC phase shift oscillator	Wein bridge oscillator
1.	This is an RC oscillator used to generate sinusoidal oscillations in the audio frequency range.	This is also an RC oscillator used to generate sinusoidal oscillations in the audio frequency range.
2.	The feedback network contains three identical RC sections.	The feedback network contains lead-lag network.
3.	Each RC section is designed to introduce 60° phases shift at the oscillator frequency. The total phase shift in the feedback network is 180° .	The feedback network introduces zero degree phase shift at the oscillator frequency.
4.	This circuit uses inverting amplifier which introduces a phase shift of 180° .	This circuit uses non-inverting amplifier with no phase shift.
5.	The frequency of oscillation is $f = \frac{1}{2\pi RC \sqrt{6+4k}}$	The frequency of oscillation is $f = \frac{1}{2\pi RC}$
6.	Condition for sustained oscillations is $h_{fe} > 4k + 23 + 29/k$	Condition for sustained oscillations is $A \geq 3$
7.	Variable frequency operation is obtained by varying all the three capacitors of the feedback network simultaneously	Variable frequency operation is obtained by varying both the feedback network capacitors simultaneously

◆ 8.8 LC OSCILLATORS (TUNED OSCILLATORS)

LC oscillators employ parallel LC circuit to generate sinusoidal oscillations. Parallel LC circuit is also called tuned circuit or resonant circuit. The frequency of oscillations is determined from the resonant condition of the tuned circuit. These oscillators exhibit high Q than RC oscillators resulting in good frequency stability. They use relatively small reactive elements in the tank circuit. They are used to generate oscillations in the frequency range from 100 kHz to hundreds of megahertz.

Figure 8.8 shows the basic configuration of LC oscillators. Based on the nature of the reactive elements X_1 , X_2 and X_3 two types of LC oscillators are obtained as indicated in Table 8.1. The amplifier can be an FET amplifier, BJT amplifier or an Op-Amp amplifier.

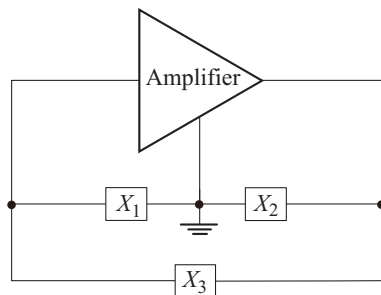


Fig. 8.8 Basic configuration of LC oscillators

Table 8.1

Oscillator type	Reactive element		
	X_1	X_2	X_3
Colpitts oscillator	C	C	L
Hartley oscillator	L	L	C

◆ 8.9 TRANSISTOR COLPITTS OSCILLATOR

Figure 8.9 shows the transistor colpitts oscillator. It consists of a CE amplifier which introduces a phase shift of 180° . Resistors R_1 , R_2 and R_E are used to establish the desired operating point

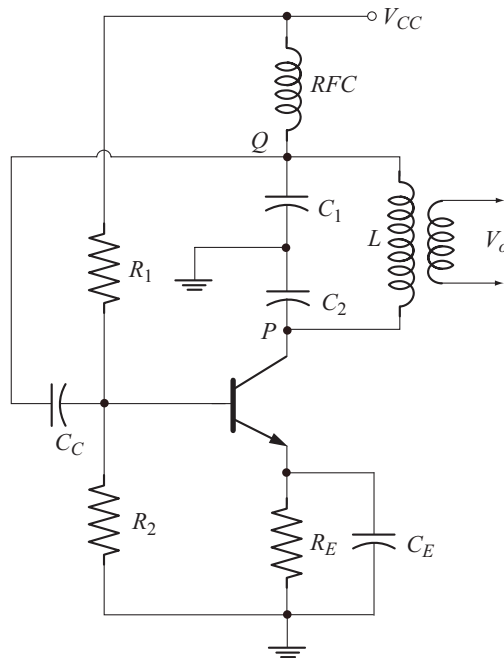


Fig. 8.9 Transistor colpitts oscillator

The feedback network consists of a tapped capacitive voltage divider C_1 and C_2 in parallel with the inductor L . The voltage across C_1 is fed back to the amplifier input through the coupling capacitor C_C .

RF choke is a large inductor. Its function is two fold:

- It acts as a dc short to the power supply V_{CC} i.e., it allows the dc current to easily pass through.
- It acts as an open circuit for ac. This ensures that the top end of the tank circuit is not connected to the ac ground.

Due to split capacitor arrangement, the tank circuit introduces 180° phase shift. As a result the oscillations at points P and Q are 180° out of phase. The total phase shift around the loop is 360° since the transistor amplifier introduces an additional phase shift of 180° .

The output is coupled to the load through a transformer. Transformer coupling has the following advantages:

- It provides electrical isolation between the oscillator output and load.
- It provides impedance matching between the oscillator output and the load.

The frequency of oscillations is given by

$$f = \frac{1}{2\pi\sqrt{LC_{eq}}} \quad (8.17)$$

$$\text{where } C_{eq} = \frac{C_1 C_2}{C_1 + C_2} \quad (8.18)$$

The condition for sustained oscillations is

$$h_{fe} \geq \frac{C_2}{C_1} \quad (8.19)$$

Example 8.12

The following data are available for the colpitts oscillator of Fig 8.9.

$$\begin{array}{lll} C_1 = 1 \text{ nF} & C_2 = 99 \text{ nF} & L = 1.5 \text{ mH} \\ L_{RFC} = 0.5 \text{ mH} & C_c = 10 \text{ }\mu\text{F} & h_{fe} = 110 \end{array}$$

- Calculate the frequency of oscillation
- Check to make sure that the condition for oscillation is satisfied.

Solution

At the oscillator frequency L_{RFC} acts as an open circuit and C_c acts as short circuit. Hence they do not appear in any of the calculations.

$$\begin{aligned} \text{(a)} \quad f &= \frac{1}{2\pi\sqrt{LC_{eq}}} \\ C_{eq} &= \frac{C_1 C_2}{C_1 + C_2} = \frac{(1 \text{ nF})(99 \text{ nF})}{1 \text{ nF} + 99 \text{ nF}} = 0.99 \text{ nF} \\ f &= \frac{1}{2\pi\sqrt{(1.5 \text{ mH})(0.99 \text{ nF})}} = 130.6 \text{ kHz} \end{aligned}$$

(b) Condition for sustained oscillation is

$$\begin{aligned} h_{fe} &\geq \frac{C_2}{C_1} \\ h_{fe} &= 110 \end{aligned}$$

$$\frac{C_2}{C_1} = \frac{99 \text{ nF}}{1 \text{ nF}} = 99$$

Note that $h_{fe} > \frac{C_2}{C_1}$

The condition for oscillation is satisfied.

Example 8.13

In a transistor colpitts oscillator $C_1 = 1 \text{ nF}$ and $C_2 = 100 \text{ nF}$. Find the value of L for a frequency of 100 kHz .

Solution

$$f = \frac{1}{2\pi\sqrt{LC_{eq}}}$$

$$\Rightarrow L = \frac{1}{[2\pi f]^2 C_{eq}}$$

$$C_{eq} = \frac{C_1 C_2}{C_1 + C_2} = \frac{(1 \text{ nF})(100 \text{ nF})}{1 \text{ nF} + 100 \text{ nF}} = 0.99 \text{ nF}$$

$$L = \frac{1}{[2\pi \times 100 \text{ kHz}]^2 [0.99 \text{ nF}]} = 2.55 \text{ mH}$$

Example 8.14

In a transistor colpitts oscillator $L = 1 \text{ mH}$, $h_{fe} = 150$. Find the values of C_1 and C_2 required for a frequency of oscillation of 120 kHz .

Solution

$$f = \frac{1}{2\pi\sqrt{LC_{eq}}}$$

$$\Rightarrow C_{eq} = \frac{1}{[2\pi f]^2 L} = \frac{1}{[2\pi \times 120 \text{ kHz}]^2 [1 \text{ mH}]} = 1.75 \text{ nF}$$

$$C_{eq} = \frac{C_1 C_2}{C_1 + C_2} \quad (\text{A})$$

The condition for sustained oscillation is

$$h_{fe} \geq \frac{C_2}{C_1}$$

$$150 \geq \frac{C_2}{C_1}$$

or $C_2 \leq 150 C_1$

$$\text{Let } C_2 = 100 C_1 \quad (\text{B})$$

Using this relation in Equation (A) we have

$$\begin{aligned} C_{eq} &= \frac{100 C_1^2}{101 C_1} \\ \Rightarrow C_1 &= \frac{101}{100} C_{eq} = \left(\frac{101}{100} \right) (1.75 \text{ nF}) \\ C_1 &= 1.7675 \text{ nF} \end{aligned}$$

From Equation (B)

$$\begin{aligned} C_2 &= (100) (1.7675) \\ C_2 &= 176.75 \text{ nF} \end{aligned}$$

Example 8.15

A colpitts oscillator uses a transistor with $h_{fe} = 120$. Find the values of C_1 , C_2 and L for a frequency of oscillation of 150 kHz.

Solution

Selection of C_1 and C_2

The smaller capacitor C_1 should be larger than the transistor parasitic capacitances and the stray wiring capacitances.

The parasitic and stray capacitances are in the order of few tens of pico Farad.

$$\begin{aligned} \text{Let } C_1 &= 1000 \text{ pF} = 1 \text{ nF} \\ h_{fe} &\geq \frac{C_2}{C_1} \\ C_2 &\leq h_{fe} C_1 \\ C_2 &\leq 120 C_1 \\ \text{Let } C_2 &= 100 C_1 \\ C_2 &= (100) (1 \text{ nF}) = 100 \text{ nF} \end{aligned}$$

Calculation of L

$$\begin{aligned} f &= \frac{1}{2\pi\sqrt{LC_{eq}}} \\ \Rightarrow L &= \frac{1}{[2\pi f]^2 C_{eq}} \\ C_{eq} &= \frac{C_1 C_2}{C_1 + C_2} = \frac{(1 \text{ nF})(100 \text{ nF})}{101 \text{ nF}} = 0.99 \text{ nF} \\ \text{Now } L &= \frac{1}{[2\pi \times 150 \times 10^3]^2 [0.99 \times 10^{-9}]} = 1.137 \text{ mH} \end{aligned}$$

Example 8.16

For the transistor colpitts oscillator derive the expressions for

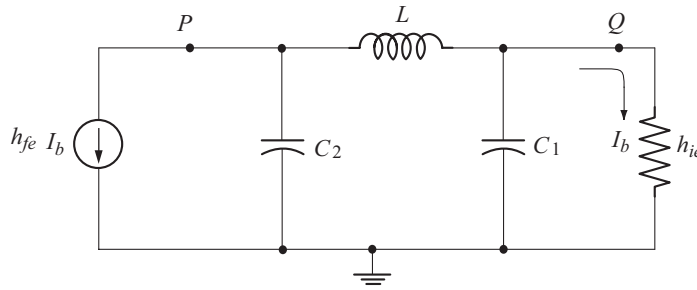
- Frequency of oscillations
- Condition for sustained oscillations

Solution

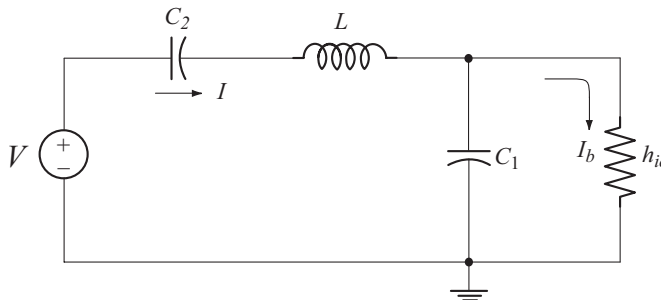
The small signal ac equivalent circuit on the output side of the colpitts oscillator of Fig. 8.9 is shown in Fig. A. The description for the construction of the equivalent circuit is given below.

- Inductor L appears between points P (collector) and Q .
- Point Q is returned to ground via the input resistance h_{ie} of the CE stage.
- $\frac{1}{h_{oe}}$ is treated as an open circuit assuming $h_{oe} \approx 0$.

RFC acts as an open circuit and C_C as short at the oscillator frequency.

**Fig. A**

Let us convert the current source in parallel with C_2 into its equivalent voltage source. The resulting circuit is shown in Fig. B.

**Fig. B**

$$V = -h_{fe} I_b \left[\frac{1}{j\omega C_2} \right] \quad (\text{A})$$

The negative sign is due to the fact that, current driven by the voltage source is upward but the actual direction of current in the current source is downward.

Using current division rule in the circuit of Fig. B, we have

$$I_b = \frac{I \left[\frac{1}{j\omega C_1} \right]}{\frac{1}{j\omega C_1} + h_{ie}}$$

$$I_b = \frac{I}{1 + j\omega C_1 h_{ie}} \quad (B)$$

Let

$$Z = \frac{1}{j\omega C_1} \parallel h_{ie}$$

$$= \frac{\left[\frac{1}{j\omega C_1} \right] h_{ie}}{\frac{1}{j\omega C_1} + h_{ie}}$$

$$Z = \frac{h_{ie}}{1 + j\omega C_1 h_{ie}} \quad (C)$$

The simplified circuit is shown in Fig. C.

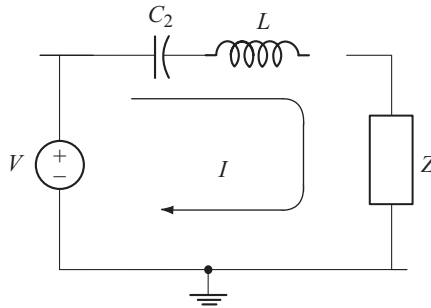


Fig. C

$$I = \frac{V}{\frac{1}{j\omega C_2} + j\omega L + Z} = \frac{V}{\frac{1 - \omega^2 LC_2}{j\omega C_2} + Z}$$

Substituting for V and Z we have

$$I = \frac{-h_{fe} I_b \left[\frac{1}{j\omega C_2} \right]}{\frac{1 - \omega^2 LC_2}{j\omega C_2} + \frac{h_{ie}}{1 + j\omega C_1 h_{ie}}}$$

$$I = \frac{-h_{fe} I_b [1 + j\omega C_1 h_{ie}]}{[1 - \omega^2 L C_2] [1 + j\omega C_1 h_{ie}] + j\omega C_2 h_{ie}} \quad (D)$$

$$\begin{aligned} Dr &= 1 - \omega^2 L C_2 + j\omega C_1 h_{ie} - j\omega^3 L C_1 C_2 h_{ie} + j\omega C_2 h_{ie} \\ &= [1 - \omega^2 L C_2] + jh_{ie} [\omega C_1 + \omega C_2 - \omega^3 L C_1 C_2] \end{aligned}$$

$$\text{Now, } I = \frac{-h_{fe} I_b [1 + j\omega C_1 h_{ie}]}{[1 - \omega^2 L C_2] + jh_{ie} [\omega C_1 + \omega C_2 - \omega^3 L C_1 C_2]} \quad (E)$$

Substituting this relation in equation (B) we have

$$\begin{aligned} I_b &= \left[\frac{1}{1 + j\omega C_1 h_{ie}} \right] \left[\frac{-h_{fe} I_b [1 + j\omega C_1 h_{ie}]}{[1 - \omega^2 L C_2] + jh_{ie} [\omega C_1 + \omega C_2 - \omega^3 L C_1 C_2]} \right] \\ 1 &= \frac{h_{fe}}{[\omega^2 L C_2 - 1] - jh_{ie} [\omega C_1 + \omega C_2 - \omega^3 L C_1 C_2]} \end{aligned} \quad (F)$$

Since LHS is real, the right hand side should also be real. This requires that

$$h_{ie} [\omega C_1 + \omega C_2 - \omega^3 L C_1 C_2] = 0 \quad (G)$$

Since $h_{ie} \neq 0$, we have

$$\begin{aligned} \omega C_1 + \omega C_2 - \omega^3 L C_1 C_2 &= 0 \\ \omega [C_1 + C_2] &= \omega^3 L C_1 C_2 \\ 1 &= \omega^2 L \frac{C_1 C_2}{C_1 + C_2} \end{aligned} \quad (H)$$

$$\text{Let } C_{eq} = \frac{C_1 C_2}{C_1 + C_2} \quad (I)$$

Using this relation in equation (H) we have

$$\begin{aligned} 1 &= \omega^2 L C_{eq} \\ \text{or } \omega &= \frac{1}{\sqrt{L C_{eq}}} \end{aligned} \quad (J)$$

$$\text{or } f = \frac{1}{2\pi \sqrt{L C_{eq}}} \quad (K)$$

Using Equation (G) in Equation (F) we have

$$\begin{aligned} 1 &= \frac{h_{fe}}{\omega^2 L C_2 - 1} \\ h_{fe} &= \omega^2 L C_2 - 1 \end{aligned} \quad (L)$$

From Equation (H)

$$\omega^2 L C_2 = \frac{C_1 + C_2}{C_1} = 1 + \frac{C_2}{C_1}$$

$$\therefore \omega^2 L C_2 - 1 = \frac{C_2}{C_1}$$

Now Equation (L) becomes

$$h_{fe} = \frac{C_2}{C_1}$$

In practical oscillators, to ensure sustained oscillations the ratio $\frac{C_2}{C_1}$ is kept smaller than h_{fe}

$$\therefore h_{fe} \geq \frac{C_2}{C_1} \quad (M)$$

◆ 8.10 TRANSISTOR HARTLEY OSCILLATOR

Figure. 8.10 shows the transistor Hartley oscillator. It consists of a CE amplifier which introduces a phase shift of 180° . Resistors R_1 , R_2 and R_E are used to establish the desired operating point.

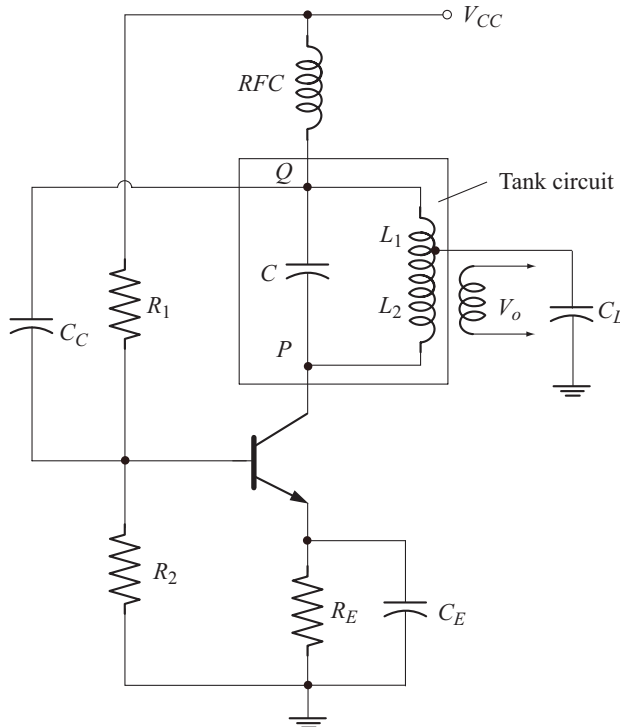


Fig. 8.10 Transistor Hartley oscillator

The feedback network consists of a tapped inductive voltage divider L_1 and L_2 in parallel with the capacitor C . The voltage across L_1 is fed back to the amplifier input through the coupling capacitor C_C .

RF choke is a large inductor. It serves two functions.

- It acts as a dc short to the power supply V_{CC} i.e., it allows the dc current to easily pass through.
- It acts as an open circuit for ac. This ensures that the top end of the tank circuit is not connected to the ac ground.

The capacitor C_L creates ac ground at the junction of L_1 and L_2 .

Due to split inductor arrangement, the tank circuit introduces 180° phase shift. As a result the oscillations at points P and Q are 180° out of phase. The total phase shift around the loop is 360° , since the transistor amplifier introduces an additional phase shift of 180° .

The output is coupled to the load through a transformer. Transformer coupling has the following advantages:

- It provides electrical isolation between the oscillator output and the load.
- It provides impedance matching between the oscillator output and the load.

The frequency of oscillations is given by

$$f = \frac{1}{2\pi\sqrt{L_{eq}C}} \quad (8.20)$$

$$\text{where } L_{eq} = L_1 + L_2 + 2M \quad (8.21)$$

M is the mutual inductance between L_1 and L_2 .

The condition for sustained oscillation is

$$h_{fe} \geq \frac{L_1 + M}{L_2 + M} \quad (8.22)$$

Neglecting the mutual inductance, the condition for sustained oscillations is given by

$$h_{fe} \geq \frac{L_1}{L_2} \quad (8.23)$$

Example 8.17

The following circuit parameter values are given for the Hartley oscillator of Fig. 8.10.

$$\begin{array}{lll} L_1 = 750 \mu\text{H} & L_2 = 750 \mu\text{H} & M = 150 \mu\text{H} \\ L_{RFC} = 0.5 \text{ mH} & C = 150 \text{ pF} & C_L = 10 \mu\text{F} \quad h_{fe} = 50 \end{array}$$

- Calculate the frequency of oscillations.
- Check to make sure that the condition for oscillation is satisfied.

Solution

$$(a) \quad f = \frac{1}{2\pi\sqrt{L_{eq}C}}$$

$$\begin{aligned}
 L_{eq} &= L_1 + L_2 + 2M \\
 &= 750 \mu\text{H} + 750 \mu\text{H} + (2)(150 \mu\text{H}) = 1800 \mu\text{H} \\
 f &= \frac{1}{2\pi\sqrt{(1800 \mu\text{H})(150 \text{ pF})}} = 306.29 \text{ kHz}.
 \end{aligned}$$

(b) Condition for sustained oscillation taking mutual inductance into account is

$$h_{fe} \geq \frac{L_1 + M}{L_2 + M}$$

$$h_{fe} = 50$$

$$\frac{L_1 + M}{L_2 + M} = \frac{750 \mu\text{H} + 150 \mu\text{H}}{750 \mu\text{H} + 150 \mu\text{H}} = 1$$

$$\text{Note that } h_{fe} > \frac{L_1 + M}{L_2 + M}$$

$\therefore$ The condition for oscillation is satisfied.

Example 8.18

In a transistor Hartley oscillator, $L_1 = 10 \mu\text{H}$, $L_2 = 10 \mu\text{H}$. Find the value of C required for an oscillating frequency of 150 kHz.

Solution

$$\begin{aligned}
 f &= \frac{1}{2\pi\sqrt{L_{eq}C}} \\
 \Rightarrow C &= \frac{1}{[2\pi f]^2 L_{eq}} \\
 L_{eq} &= L_1 + L_2 + 2M
 \end{aligned}$$

Since M is not given we can take $M = 0$

$$\begin{aligned}
 L_{eq} &= 10 \mu\text{H} + 10 \mu\text{H} = 20 \mu\text{H} \\
 C &= \frac{1}{[2\pi \times 150 \times 10^3]^2 [20 \mu\text{H}]} = 56.28 \text{ nF}
 \end{aligned}$$

Example 8.19

In a transistor Hartley oscillator $C = 0.01 \mu\text{F}$ and $h_{fe} = 50$. Find the values of L_1 and L_2 required for a frequency of oscillation of 150 kHz.

Solution

$$f = \frac{1}{2\pi\sqrt{L_{eq}C}}$$

$$\begin{aligned}
 L_{eq} &= \frac{1}{[2\pi f]^2 C} \\
 &= \frac{1}{[2\pi \times 150 \times 10^3]^2 [0.01 \times 10^{-6}]} = 112.5 \mu\text{H}
 \end{aligned}$$

Neglecting mutual inductance

$$\begin{aligned}
 L_{eq} &= L_1 + L_2 \\
 \therefore L_1 + L_2 &= 112.5 \mu\text{H} \quad (\text{A})
 \end{aligned}$$

Condition for oscillation to start is

$$\begin{aligned}
 h_{fe} &\geq \frac{L_1}{L_2} \\
 50 &\geq \frac{L_1}{L_2} \\
 \frac{L_1}{L_2} &\leq 50 \\
 \text{Let } \frac{L_1}{L_2} &= 10 \\
 \Rightarrow L_1 &= 10 L_2 \quad (\text{B})
 \end{aligned}$$

Using this relation in Equation (A) we have

$$\begin{aligned}
 11 L_2 &= 112.5 \mu\text{H} \\
 L_2 &= 10.23 \mu\text{H}
 \end{aligned}$$

From Equation (B)

$$L_1 = 102.3 \mu\text{H}$$

Example 8.20

A Hartley oscillator uses a transistor with $h_{fe} = 40$. Find the values of L_1 , L_2 and C for a frequency of oscillations of 100 kHz.

Solution

Selection of C

Select C Larger than parasitic and stray capacitances

$$\text{Let } C = 10 \text{ nF} = 0.01 \mu\text{F}$$

Calculation of L_1 and L_2

$$L_{eq} = \frac{1}{[2\pi f]^2 C}$$

$$= \frac{1}{\left[2\pi \times 100 \times 10^3\right]^2 \left[0.01 \times 10^{-6}\right]} = 253.3 \mu\text{H}$$

Neglecting mutual inductance

$$\begin{aligned} L_{eq} &= L_1 + L_2 \\ \therefore L_1 + L_2 &= 253.3 \mu\text{H} \end{aligned} \quad (\text{A})$$

Condition for sustained oscillations is

$$\begin{aligned} h_{fe} &\geq \frac{L_1}{L_2} \\ 40 &\geq \frac{L_1}{L_2} \\ \text{or } \frac{L_1}{L_2} &\leq 40 \\ \text{Let } \frac{L_1}{L_2} &= 10 \\ \Rightarrow L_1 &= 10 L_2 \end{aligned} \quad (\text{B})$$

Using this relation in Equation (A) we have

$$\begin{aligned} 11 L_2 &= 253.3 \mu\text{H} \\ L_2 &= 23.02 \mu\text{H} \end{aligned}$$

From equation (B)

$$\begin{aligned} L_1 &= 10 L_2 = (10) (23.02 \mu\text{H}) \\ L_1 &= 230.2 \mu\text{H} \end{aligned}$$

Example 8.21

For the transistor Hartley oscillator derive the expressions for

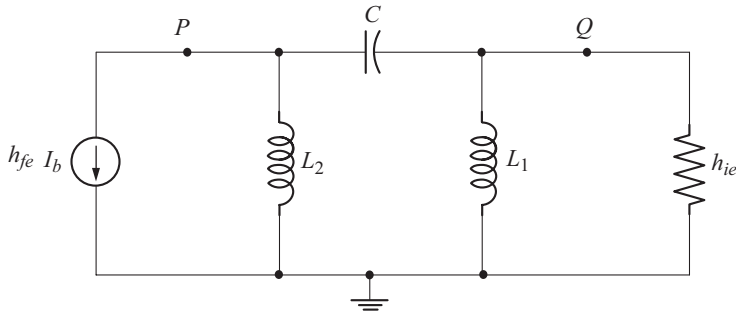
- Frequency of oscillation
- Condition for sustained oscillation

Solution

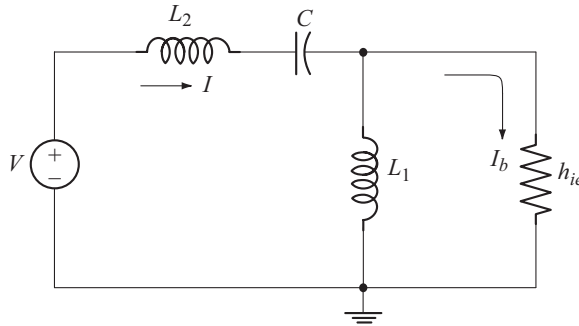
The small signal ac equivalent circuit on the output side of the Hartley oscillator of Fig. 8.10 is shown in Fig. A.

The description for the construction of equivalent circuit is given below:

- Capacitor C appears between points P (collector) and Q .
- Point Q is returned to the ground via the input resistance h_{ie} of the CE stage.
- $\frac{1}{h_{oe}}$ is treated as an open circuit and C_L as short circuit at the oscillator frequency.
- Mutual Inductance between L_1 and L_2 is neglected.

**Fig. A**

Let us convert the current source in parallel with L_2 into its equivalent voltage source as shown in Fig. B.

**Fig. B**

$$V = -h_{fe} I_b [j\omega L_2] \quad (A)$$

The negative sign is due to the fact that, current driven by the voltage source is upward but the actual direction of current in the current source is downward.

Using current division rule in the circuit of Fig. B, we get

$$I_b = \frac{I [j\omega L_1]}{h_{ie} + j\omega L_1} \quad (B)$$

Let

$$Z = j\omega L_1 \parallel h_{ie}$$

$$Z = \frac{[j\omega L_1] h_{ie}}{h_{ie} + j\omega L_1} \quad (C)$$

The simplified circuit is shown in Fig. C.

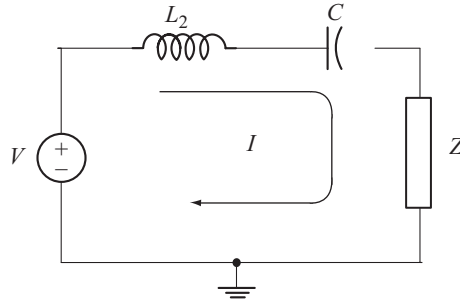


Fig. C

$$I = \frac{V}{j\omega L_2 + \frac{1}{j\omega C} + Z} = \frac{V}{\frac{1 - \omega^2 L_2 C}{j\omega C} + Z}$$

Substituting for V and Z we have

$$I = \frac{-h_{fe} I_b [j\omega L_2]}{\frac{1 - \omega^2 L_2 C}{j\omega C} + \frac{j\omega L_1 h_{ie}}{h_{ie} + j\omega L_1}}$$

$$I = \frac{[-h_{fe} I_b] [j\omega L_2] [h_{ie} + j\omega L_1] [j\omega C]}{[1 - \omega^2 L_2 C] [h_{ie} + j\omega L_1] + [j\omega L_1 h_{ie}] [j\omega C]} \quad (D)$$

$$\begin{aligned} \text{Dr} &= [1 - \omega^2 L_2 C] h_{ie} + j\omega L_1 [1 - \omega^2 L_2 C] - \omega^2 L_1 C h_{ie} \\ &= h_{ie} [1 - \omega^2 L_2 C - \omega^2 L_1 C] + j\omega L_1 [1 - \omega^2 L_2 C] \end{aligned} \quad (E)$$

Substituting this relation in Equation (D) we get

$$I = \frac{[h_{fe} I_b] [\omega^2 L_2 C] [h_{ie} + j\omega L_1]}{h_{ie} [1 - \omega^2 L_2 C - \omega^2 L_1 C] + j\omega L_1 [1 - \omega^2 L_2 C]}$$

Substituting this relation in equation (B) we have

$$I_b = \left[\frac{j\omega L_1}{h_{ie} + j\omega L_1} \right] \left\{ \frac{[h_{fe} I_b] [\omega^2 L_2 C] [h_{ie} + j\omega L_1]}{h_{ie} [1 - \omega^2 L_2 C - \omega^2 L_1 C] + j\omega L_1 [1 - \omega^2 L_2 C]} \right\}$$

$$1 = \frac{j h_{fe} \omega^3 L_1 L_2 C}{h_{ie} [1 - \omega^2 L_2 C - \omega^2 L_1 C] + j\omega L_1 [1 - \omega^2 L_2 C]}$$

Eliminating j from the numerator we have

$$1 = \frac{h_{fe} \omega^3 L_1 L_2 C}{-j h_{ie} [1 - \omega^2 L_2 C - \omega^2 L_1 C] + \omega L_1 [1 - \omega^2 L_2 C]} \quad (F)$$

Since the LHS is real, the RHS should also be real.

This requires that

$$h_{ie} [1 - \omega^2 L_2 C - \omega^2 L_1 C] = 0 \quad (G)$$

Since $h_{ie} \neq 0$, we have

$$\begin{aligned} 1 - \omega^2 L_2 C - \omega^2 L_1 C &= 0 \\ \omega^2 C [L_1 + L_2] &= 1 \end{aligned} \quad (H)$$

$$\text{Let } L_{eq} = L_1 + L_2 \quad (I)$$

$$\text{Now } \omega^2 C L_{eq} = 1$$

$$\text{or } \omega = \frac{1}{\sqrt{L_{eq} C}}$$

$$\text{or } f = \frac{1}{2\pi\sqrt{L_{eq} C}} \quad (J)$$

Condition for sustained oscillations

Substituting Equation (G) in Equation (F) we get

$$\text{or } 1 = \frac{h_{fe} \omega^3 L_1 L_2 C}{\omega L_1 [1 - \omega^2 L_2 C]} \quad (K)$$

From Equation (H)

$$\begin{aligned} \omega^2 L_1 C + \omega^2 L_2 C &= 1 \\ \Rightarrow 1 - \omega^2 L_2 C &= \omega^2 L_1 C \end{aligned}$$

Using this relation in Equation (K) we get

$$\text{or } h_{fe} = \frac{L_1}{L_2} \quad (L)$$

In practical oscillators, to ensure oscillations, the ratio $\frac{L_1}{L_2}$ is kept smaller than h_{fe}

$$\therefore h_{fe} \geq \frac{L_1}{L_2} \quad (M)$$

◆ 8.11 CRYSTAL OSCILLATOR

A crystal oscillator is basically a tuned circuit oscillator. It has the same circuit topology as the colpitts oscillator except that it uses a piezo electric crystal instead of an inductor. The crystal quartz usually has a greater stability in holding constant at whatever frequency the crystal is originally cut to operate. Crystal oscillators are used in communication transmitters and receivers where high frequency stability is required.

◆ 8.12 PIEZO ELECTRIC EFFECT

Many naturally available crystals like Rochelle salt, tourmaline, quartz etc., exhibit piezoelectric effect. Piezoelectric effect is an electromechanical phenomenon. If the crystal is mechanically vibrated it develops an AC voltage across the ends of the face of the crystal. If an AC field is applied across the face of the crystal it vibrates mechanically. The natural resonant frequency of the crystal depends upon the dimensions of the crystal and also on the mechanical orientation of the crystal structure. The resonant frequency of the crystal is inversely proportional to the thickness of the crystal. Thus the frequency of oscillations increases as the thickness of the crystal decreases.

Quartz crystals are the natural choice in sinusoidal oscillators due to the following reasons:

- They are mechanically strong.
- They have good piezoelectric sensitivity.
- They are less expensive.

◆ 8.13 CHARACTERISTICS OF QUARTZ CRYSTALS

Figure 8.11(a) shows the piezoelectric crystal mounted between the plates and connected across an AC source of variable frequency. Figure 8.11(b) shows the electrical equivalent circuit of the mechanically vibrating crystal.

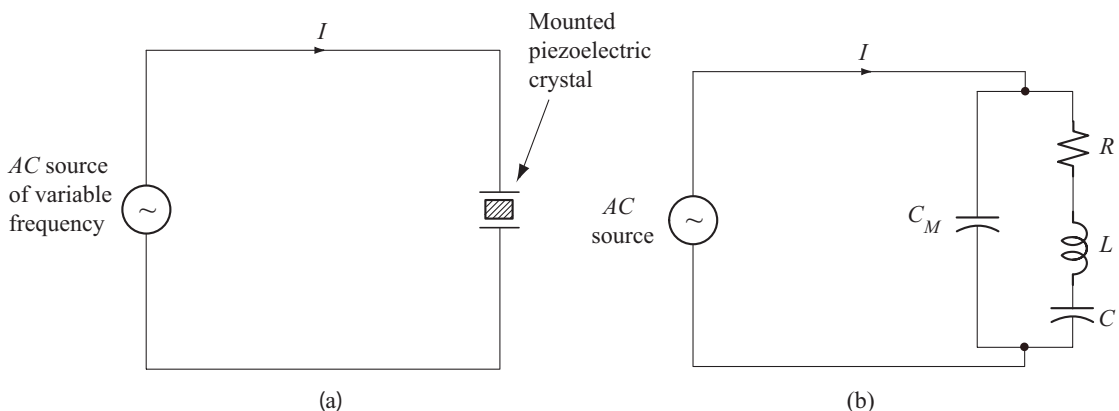


Fig. 8.11 Mounted piezoelectric crystal and its electrical equivalent circuit

- L = Electrical equivalent inductance of crystal mass.
 C = Electrical equivalent capacitance of the crystal compliance.
 R = Electrical equivalent resistance of the crystal structure's internal friction.
 C_M = Capacitance due to mechanical mounting of the crystal.

For a 2 MHz crystal the typical values of these parameters are:

$$\begin{array}{ll}
 R = 82 \, \Omega & L = 0.52 \, \text{H} \\
 C = 0.0122 \, \text{pF} & C_M = 4.27 \, \text{pF}
 \end{array}$$

This crystal has a quality factor of 80,000. Note that high quality factor implies, high frequency stability.

The frequency response of a crystal is obtained by plotting the current I through the crystal as a function of the frequency of ac source. Figure 8.12(a) shows the frequency response of the mounted crystal.

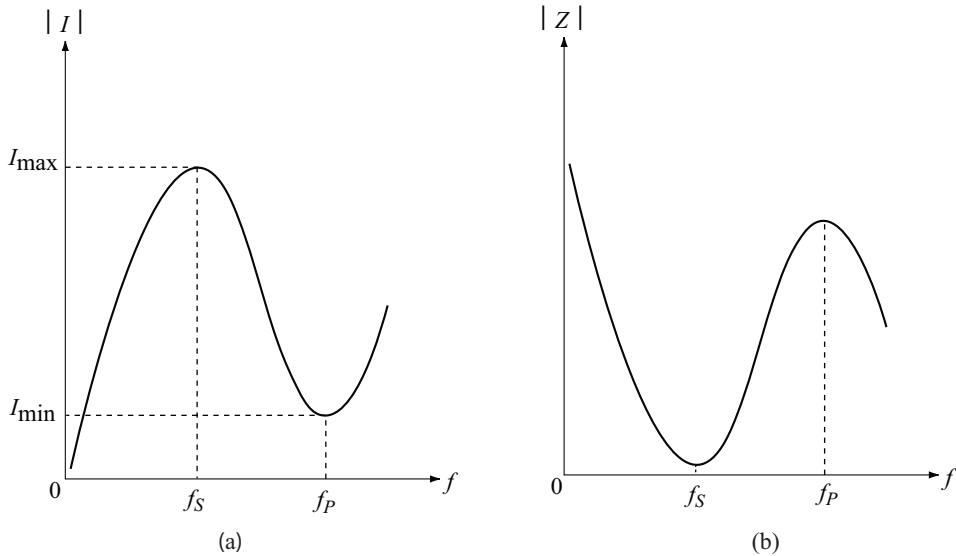


Fig. 8.12 (a) Frequency response of piezoelectric crystal
(b) Crystal impedance versus frequency

When the frequency of the ac source is equal to the frequency f_s , the current through the crystal becomes maximum ($I_{\max}$). This condition is called the series resonance and f_s is called the series resonant frequency. Series resonance occurs when the reactance of L is equal to the reactance of C in the series RLC branch.

$$\begin{aligned}
 \text{i.e.,} \quad \omega L &= \frac{1}{\omega C} \\
 \omega^2 &= \frac{1}{LC}
 \end{aligned}$$

$$\omega = \frac{1}{\sqrt{LC}}$$

$$f = f_s = \frac{1}{2\pi\sqrt{LC}} \quad (8.24)$$

Since the current is maximum, the impedance of the crystal is minimum.

When the frequency of the ac source is equal to the frequency $f_p > f_s$, the current through the crystal becomes minimum ($I_{\min}$). This condition is called parallel resonance and f_p is called the parallel resonant frequency. Parallel resonance occurs when the reactance of L is equal to the sum of the reactances of C_M and C .

i.e., $\omega L = \frac{1}{\omega C_M} + \frac{1}{\omega C}$

$$\omega^2 = \frac{1}{L} \left[\frac{1}{C_M} + \frac{1}{C} \right]$$

Taking $\frac{1}{C_p} = \frac{1}{C_M} + \frac{1}{C}$ (8.25)

or $C_p = \frac{C_M C}{C_M + C}$ (8.26)

We have, $\omega^2 = \frac{1}{LC_p}$

$$\omega = \frac{1}{\sqrt{LC_p}}$$

or $f = f_p = \frac{1}{2\pi\sqrt{LC_p}}$ (8.27)

Since the current is minimum, the impedance of the crystal is maximum.

Variation of $|Z|$ versus frequency is shown in Fig. 8.12(b).

Also since $C_M \gg C$, $\frac{1}{C_M} \ll \frac{1}{C}$

From Equation (8.25) we get

$$\frac{1}{C_p} \approx \frac{1}{C} \Rightarrow C_p \approx C$$

Now from Equation (8.27)

$$f_p \approx \frac{1}{2\pi\sqrt{LC}} = f_s \quad (8.28)$$

Note that f_p and f_s are very close to each other.

◆ 8.14 CRYSTAL REACTANCE

Figure 8.13 shows the variation of crystal reactance with frequency. Note that crystal exhibits inductive reactance only between f_s and f_p .

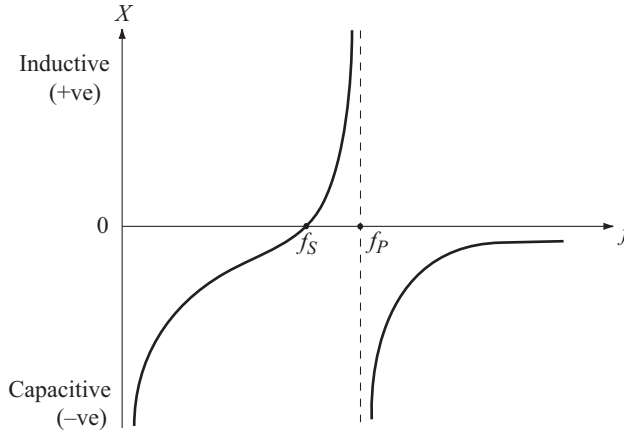


Fig. 8.13 Crystal reactance versus frequency

Since f_s is very close to f_p , the crystal reactance is inductive only at $f_s \approx f_p$. When the inductor is replaced with the crystal in the colpitts oscillator, it oscillates only at that particular frequency of resonance where the crystal reactance is inductive.

◆ 8.15 TRANSISTOR CRYSTAL OSCILLATOR

The crystal can be operated either in the series resonant mode or in the parallel resonant mode. These two modes are discussed in the following sections.

8.15.1 Crystal Oscillator in Series Resonant Mode

Figure 8.14 shows the circuit of crystal oscillator in which the crystal is operated in series resonant mode.

The circuit uses a transistor CE stage. R_1 , R_2 and R_E are selected so as to establish the desired Q point. C_E bypasses the emitter resistor R_E . The coupling capacitor C_c is selected such that it acts as a short circuit at the oscillator frequency. RFC acts as a short for dc current and open circuit for ac signal. Thus it prevents the oscillations from reaching the dc supply V_{CC} .

Crystal is used as series element in the feedback path, so that it operates in series resonant mode. Crystal has minimum impedance at the series resonant frequency f_s . Hence maximum feedback from collector to base occurs at this frequency. Thus oscillations are sustained at the series resonant frequency f_s .

Once the oscillations are set up the frequency of oscillations is held stabilized at f_s by the crystal. Changes in supply voltage, transistor device parameters etc have no effect on the circuit operating frequency. The frequency stability of the circuit is set by the frequency stability of the crystal, which is very high.

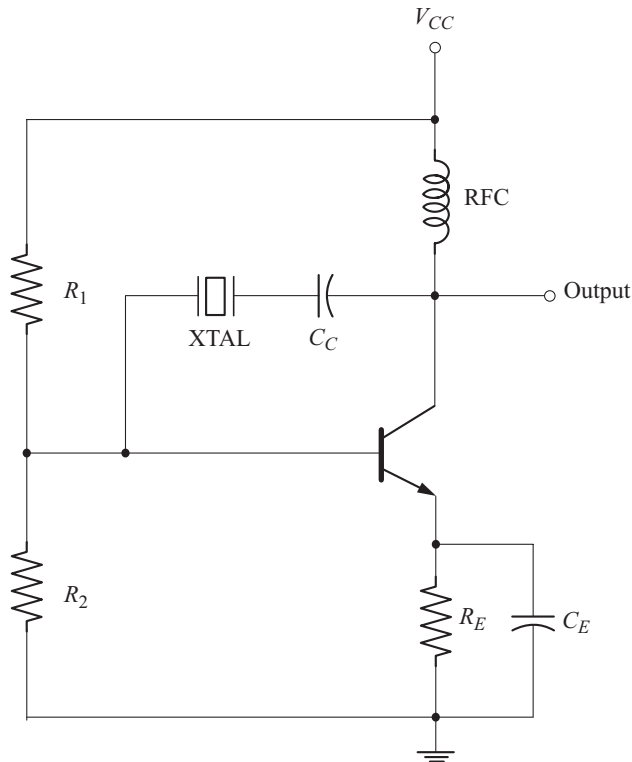


Fig. 8.14 Transistor crystal oscillator operating in series resonant mode

8.15.2 Crystal Oscillator in Parallel Resonant Mode

Figure 8.15 shows the circuit of crystal oscillator in which the crystal is operated in the parallel-resonant mode.

The purpose of R_1 , R_2 , R_E and RFC has been already explained in the previous section. Crystal is used as shunt element in the tank circuit so that it operates in the parallel resonant mode. Note that the tank circuit is same as that of colpitts oscillator with inductor replaced by the crystal.

Maximum voltage is developed across the crystal at its parallel resonant frequency since it has maximum impedance at this frequency. Also the crystal behaves as an inductor at its parallel resonant frequency. The crystal along with C_1 and C_2 gives the tank circuit effect at this frequency. Thus oscillations are sustained at the parallel resonant frequency f_p . Series combination of C_1 and C_2 acts as a voltage divider for the output voltage. The voltage across C_2 is fed to the emitter of the transistor. Note that the feedback voltage is maximum at the parallel resonant frequency of the crystal.

C_B is the bypass capacitor, which acts as a short circuit at the oscillator frequency, creating ac ground at the base. As a result, oscillations will not reach the base and there is no feedback at the base terminal.

The frequency of oscillations is held stabilized at f_p by the crystal. Changes in supply voltage, transistor device parameters etc have no effect on the circuit operating frequency. The frequency stability of the circuit is set by the frequency stability of the crystal, which is very high.

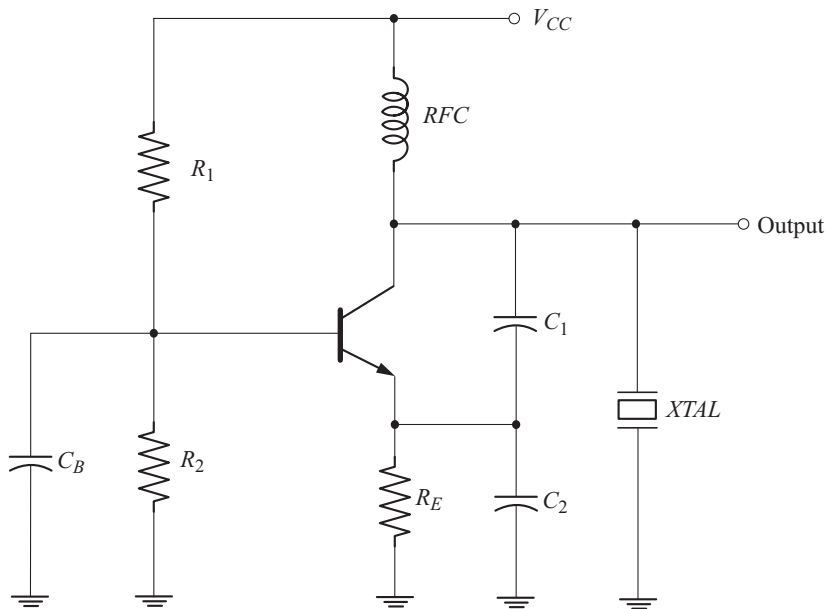


Fig. 8.15 Transistor crystal oscillator operating in parallel resonant mode

◆ 8.16 MERITS AND DEMERITS OF CRYSTAL OSCILLATOR

The merits of crystal oscillator are:

- The frequency stability of the crystal is very high. It is around one part in 10^6 i.e., 0.0001% per day. For example the frequency drift of a 1MHz crystal is 1Hz per day which is very negligible.
- Crystal with frequency upto 10 MHz can be constructed with very high frequency stability.
- The temperature stability of a crystal is very good. i.e., the frequency drift due to temperature change is negligibly small. Typically the frequency drift for 1°C rise in temperature of the crystal is around 10–12 Hz per mega hertz.
- Quartz crystals are readily available in nature. The crystal is very small in size, inexpensive and lighter in weight.
- Crystal replaces inductor in the tank circuit. As a result, crystal oscillator circuits are less bulky, inexpensive and lighter in weight.

The demerits of the crystal oscillator are:

- Crystals are very delicate and hence require careful handling.
- The thickness of the crystal is inversely proportional to the frequency. Hence higher frequency crystals are thinner in size and are mechanically weak.
- The frequency of the crystal depends on the crystal dimensions and how the crystal is cut. Hence for a given crystal, its frequency of oscillations is fixed. Whenever a new frequency of oscillation is required, the whole circuit is to be redesigned and the crystal is to be replaced.

◆ 8.17 APPLICATIONS OF CRYSTAL OSCILLATOR

Following are the some of the applications of crystal oscillators:

- To generate clock signal for computers and other synchronous digital systems.
- To generate carrier frequency in communication transmitters.
- To generate local oscillator frequency in communication receivers. Crystal kept in temperature controlled oven is used to generate highly precise clock which is used for time standards.

Example 8.22

A crystal has the following parameters:

$$\begin{aligned} L &= 0.334 \text{ H} & C &= 0.065 \text{ pF} \\ C_M &= 1 \text{ pF} & R &= 5.5 \text{ k}\Omega \end{aligned}$$

- Calculate the series resonant frequency.
- Calculate the parallel resonant frequency.
- By what percent does the parallel-resonant frequency exceed the series resonant frequency?
- Find the Q of the crystal.

Solution

$$\begin{aligned} \text{(a)} \quad f_s &= \frac{1}{2\pi\sqrt{LC}} \\ &= \frac{1}{2\pi\sqrt{(0.334)(0.065 \times 10^{-12})}} = 1.08016 \text{ MHz} \end{aligned}$$

$$\omega_s = 2\pi f_s = 6.7868 \text{ M rad/sec}$$

$$\begin{aligned} \text{(b)} \quad f_p &= \frac{1}{2\pi\sqrt{LC_p}} \\ C_p &= \frac{CC_M}{C + C_M} = \frac{(0.065 \text{ pF})(1 \text{ pF})}{0.065 \text{ pF} + 1 \text{ pF}} = 0.061 \text{ pF} \\ f_p &= \frac{1}{2\pi\sqrt{(0.334)(0.061 \times 10^{-12})}} = 1.1150 \text{ MHz} \end{aligned}$$

$$\omega_p = 2\pi f_p = 7.005 \text{ M rad/sec}$$

$$\begin{aligned} \text{(c)} \quad \% \text{ difference between } f_p \text{ and } f_s &= \frac{f_p - f_s}{f_s} \times 100\% \\ &= \frac{1.1150 - 1.08016}{1.08016} \times 100\% = 3.22\% \end{aligned}$$

f_p is more than f_s by 3.22%

$$(d) \quad Q = \frac{\omega_s L}{R} = \frac{(6.7868 \times 10^6)(0.334)}{(5.5 \times 10^3)} = 412.14$$

Example 8.23

- (a) Prove that the ratio of the parallel to series resonant frequencies is given approximately by $1 + \frac{C}{2C_M}$.
- (b) If $C = 0.04$ pF and $C_M = 2$ pF, by what percent is the parallel-resonant frequency greater than the series resonant frequency?

Solution

$$(a) \quad f_s = \frac{1}{2\pi\sqrt{LC}}$$

$$f_p = \frac{1}{2\pi\sqrt{LC_p}}$$

$$\frac{f_p}{f_s} = \sqrt{\frac{C}{C_p}}$$

But $C_p = \frac{CC_M}{C+C_M}$

Now $\frac{f_p}{f_s} = \sqrt{C \times \left[\frac{C+C_M}{C C_M} \right]}$

$$\frac{f_p}{f_s} = \left[1 + \frac{C}{C_M} \right]^{1/2} \quad (A)$$

Since $\frac{C}{C_M} \ll 1$, we can write Equation (A) using binomial expansion as

$$\frac{f_p}{f_s} \approx 1 + \frac{C}{2C_M} \quad (B)$$

(b) $C = 0.04$ pF $C_M = 2$ pF

$$\frac{f_p}{f_s} = 1 + \frac{0.04 \text{ pF}}{2 \times 2 \text{ pF}} = 1.01$$

$$f_p = 1.01 f_s \text{ or } f_p \text{ is } 101\% \text{ of } f_s$$

$$\therefore f_p \text{ is greater than } f_s \text{ by } 1\%.$$

**Exercise Problems**

- 8.1** In a transistor RC phase shift oscillator $R = 3.9 \text{ k}\Omega$ and $C = 0.01 \text{ }\mu\text{F}$. If $R_C = 4.7 \text{ k}\Omega$ calculate the frequency of oscillation.
- 8.2** Find the values of R and C for a transistor RC-phase shift oscillator for a frequency of 3 kHz. $R_C = 5 \text{ k}\Omega$ and for the transistor $h_{fe} = 100$.
- 8.3** Design the component values for a Weinbridge oscillator for a frequency of 1.5 kHz.
- 8.4** In a transistor Hartley oscillator $L_1 = L_2 = 10 \text{ mH}$, $M = 50 \text{ }\mu\text{H}$ and $C = 0.01 \text{ }\mu\text{F}$. Calculate the frequency of oscillations.
- 8.5** In a transistor Colpitt's oscillator $C_1 = C_2 = 0.01 \text{ }\mu\text{F}$, $L = 20 \text{ mH}$. Calculate the frequency of oscillation.
- 8.6** A Quartz crystal has the following data:
 $L = 0.5 \text{ H}$ $R = 1 \text{ k}\Omega$ $C = 2 \text{ pF}$ $C_M = 20 \text{ pF}$.
Calculate the following:
- (a) Series resonant frequency
 - (b) Parallel resonant frequency
 - (c) Quality factor

Chapter 9

FET AMPLIFIERS

Junction field-effect transistor (JFET) is basically a voltage controlled device. The output current of the device is controlled by the input voltage. FET has an excellent voltage gain and high input impedance. Because of high input impedance, the ac equivalent model is simpler than that of BJT which makes the ac analysis of FET amplifiers less complex. In this chapter analysis of common-source, common gate and common drain JFET amplifiers have been carried out. The analysis of DMOSFET and EMOSFET amplifiers have also been considered. A few design examples are included at the end of this chapter.

◆ 9.1 BASICS OF JUNCTION FIELD EFFECT TRANSISTOR

A Junction field-effect transistor, abbreviated as JFET or simply FET, is a three terminal semiconductor device in which the current flow is due to only one type of carriers i.e., either electrons or holes. Hence it is a unipolar device.

The name field-effect transistor is derived from the fact that the current flow is controlled by an electric field setup in the device by an externally applied voltage. Unlike BJT which is a current controlled device, the FET is a voltage controlled device.

Figure 9.1 shows the circuit symbols of n -channel and p -channel FETs. The three terminals of the FET are Drain (D), Source (S) and Gate (G) which are analogous to the collector, emitter and base of the BJT respectively. In n channel FET, the current flow is due to only electrons whereas in p -channel FET it is due to only holes.

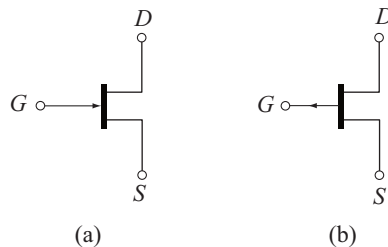


Fig. 9.1 (a) Circuit symbol of n -channel FET
(b) Circuit symbol of p -channel FET

◆ 9.2 FET CURRENT EQUATION

Figure 9.2 shows the biasing arrangement for an n -channel FET. Observe that the gate is made negative with respect to source and the drain positive with respect to source.

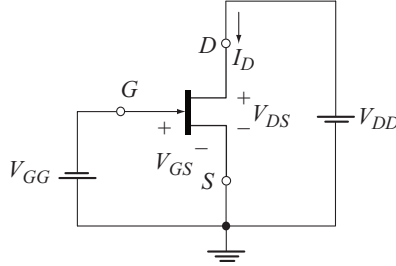


Fig. 9.2 Biasing arrangement for n -channel FET

In FET the drain current I_D is controlled by the gate-source voltage V_{GS} . The relationship between I_D and V_{GS} is defined by Shockley's equation given by

$$I_D = I_{DSS} \left[1 - \frac{V_{GS}}{V_p} \right]^2 \quad (9.1)$$

When $V_{GS} = 0$ and $V_{DS} > |V_p|$,

$$I_D = I_{DSS} \quad (9.2)$$

I_{DSS} is the maximum drain current in the FET which occurs when $V_{GS} = 0$ and $V_{DS} > |V_p|$.

When

$$\begin{aligned} V_{GS} &= V_p \\ I_D &= 0 \end{aligned}$$

V_p is called the pinch-off voltage

V_p is negative for n -channel FETs and positive for p channel FETs.

◆ 9.3 TRANSFER CHARACTERISTICS

Transfer characteristic is the plot of I_D versus V_{GS} . Figure 9.3 shows the transfer characteristics of FET. Transfer characteristics is defined by the Shockley's equation.

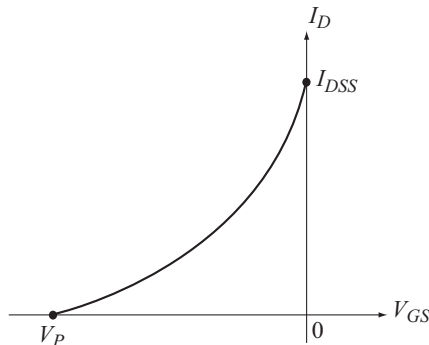


Fig. 9.3 Transfer characteristics of FET

Observe that I_D has the maximum value of I_{DSS} when $V_{GS} = 0$. It decreases for negatively increasing V_{GS} and becoming zero at $V_{GS} = V_p$. It should be noted that I_D is a non linear function of V_{GS} .

◆ 9.4 DRAIN CHARACTERISTICS

Figure 9.4 shows the drain characteristics of FET.

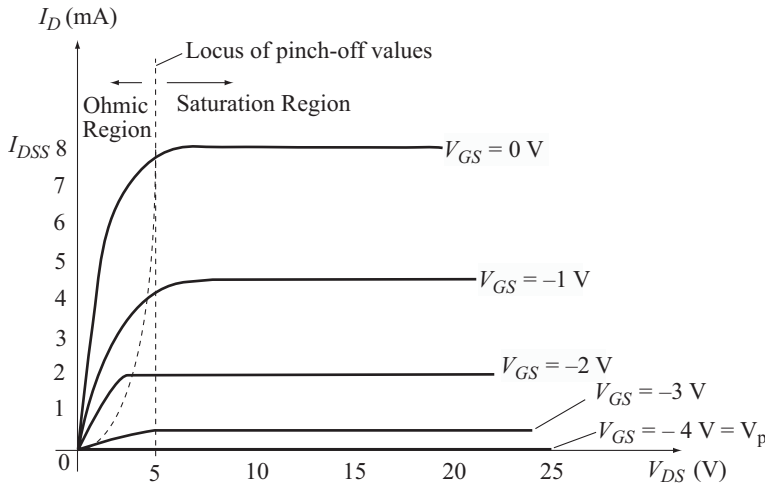


Fig. 9.4 Drain characteristics of FET

The region to the right of the pinch-off locus is called the saturation region. In this region I_D varies very little with V_{DS} but changes significantly with V_{GS} . This is the linear amplification region for the FET. When biased in this region, FET amplifies the applied signal with minimum distortion.

Therefore the biasing requirements for n -channel FET to work as a linear amplifying device are:

$$0 > V_{GS} > V_p \quad (9.3)$$

$$\text{and} \quad V_{DS} > |V_p| \quad (9.4)$$

◆ 9.5 FET SMALL-SIGNAL MODEL

The small-signal ac model of FET is required in the ac analysis of FET amplifiers. The small-signal ac model is developed using the following facts.

- The impedance between gate and source terminals of an FET is very high. Typically it is in the order of 1000 M Ω . Hence FET can be represented by an open circuit between gate and source terminals.
- The drain current is controlled by the gate-source voltage.

$$\begin{aligned} \text{i.e.,} \quad & I_d \propto V_{gs} \\ \text{or} \quad & I_d = g_m V_{gs} \end{aligned} \quad (9.5)$$

g_m is called the transconductance of FET. g_m is alternatively represented by y_{fs} . y_{fs} stands for forward transfer admittance.

As per Equation (9.5), FET can be modelled as a voltage controlled current source, $I_d = g_m V_{gs}$, between drain and source terminals.

- In the saturation region, I_d changes slightly with V_{ds}

$$\begin{aligned} \text{ie} \quad & I_d \propto V_{ds} \\ \text{or} \quad & I_d = \left[\frac{1}{r_d} \right] V_{ds} \end{aligned} \quad (9.6)$$

r_d is the output impedance of FET.

$$r_d = \frac{1}{y_{os}} \quad (9.7)$$

y_{os} is the output admittance.

As per Equation (9.6), FET can be modelled as a resistor r_d , connected between the drain and source terminals. Typically r_d lies in the range 20 k Ω – 100 k Ω .

The lower case suffixes are used to imply that I_d , V_{gs} , V_{ds} , g_m and r_d are all ac values.

Figure 9.5 shows the small-signal ac model of FET. This model applies for both n -channel p -channel and FETs.

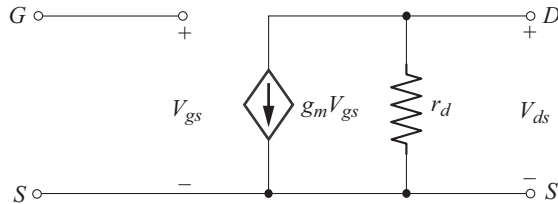


Fig. 9.5 Small signal ac model of FET

Example 9.1

An FET has $y_{fs} = 4$ mS and $y_{os} = 33.33$ μ S

- Find g_m and r_d
- Sketch the small-signal ac model of FET

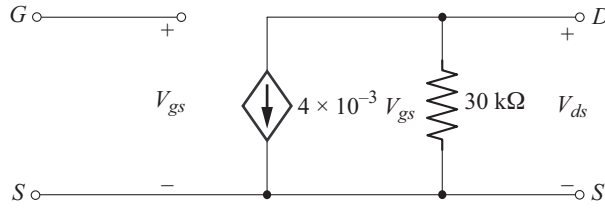
Solution

- $$g_m = y_{fs} = 4 \text{ mS}$$

$$r_d = \frac{1}{y_{os}} = \frac{1}{33.33 \mu\text{S}}$$

$$r_d = 30 \text{ k}\Omega$$

- The small-signal ac model is shown below.



◆ 9.6 GRAPHICAL DETERMINATION OF g_m

The transconductance g_m is given by the slope of the transfer characteristics at the Q point.

$$g_m = \left. \frac{\Delta I_D}{\Delta V_{GS}} \right|_{Q \text{ point}} \quad (9.8)$$

Graphical determination of g_m is illustrated in Fig. 9.6.

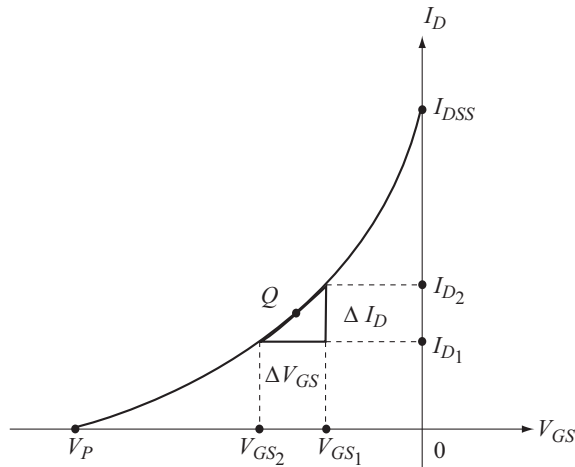


Fig. 9.6 Graphical determination of g_m

From Fig. 9.6 we have

$$\begin{aligned} \Delta I_D &= I_{D2} - I_{D1} \\ \Delta V_{GS} &= V_{GS2} - V_{GS1} \end{aligned}$$

$$\text{Now} \quad g_m = \frac{I_{D2} - I_{D1}}{V_{GS2} - V_{GS1}} \quad (9.9)$$

◆ 9.7 MATHEMATICAL EXPRESSION FOR g_m

g_m is given by the slope of the transfer characteristics at the Q point. But slope of transfer characteristics is given by the derivative of I_D with respect to V_{GS} at the Q point.

$$\therefore \quad g_m = \left. \frac{dI_D}{dV_{GS}} \right|_{Q\text{-point}} \quad (9.10)$$

$$\text{But} \quad I_D = I_{DSS} \left[1 - \frac{V_{GS}}{V_p} \right]^2$$

$$\text{Now} \quad g_m = 2 I_{DSS} \left[1 - \frac{V_{GS}}{V_p} \right] \left[\frac{-1}{V_p} \right] \quad (9.11)$$

In Equation (9.11), $\frac{V_{GS}}{V_p}$ is positive, for both n -channel and p -channel FETs. To get positive value for g_m , let us omit negative sign and write $|V_p|$ instead of V_p .

$$\text{Hence} \quad g_m = \frac{2I_{DSS}}{|V_p|} \left[1 - \frac{V_{GS}}{V_p} \right] \quad (9.12)$$

When $V_{GS} = 0$,

$$g_m = \frac{2I_{DSS}}{|V_p|}$$

Note that this is the highest possible value of g_m .

$$\text{Let} \quad g_{m_o} = \frac{2I_{DSS}}{|V_p|} \quad (9.13)$$

Using Equation (9.13) in Equation (9.12) we have

$$g_m = g_{m_o} \left[1 - \frac{V_{GS}}{V_p} \right] \quad (9.14)$$

Observe from Equation (9.14) that, g_m decreases from g_{m_o} to 0 when V_{GS} is increased negatively from 0 to V_p .

From Equation (9.14) we have

$$g_m = \left[-\frac{g_{m_o}}{V_p} \right] V_{GS} + g_{m_o} \quad (9.15)$$

Equation (9.15) represents a straight line in $V_{GS} - g_m$ plane with slope $-\frac{g_{m_o}}{V_p}$ and intercept g_{m_o} . The plot of g_m versus V_{GS} is shown in Fig. 9.7.

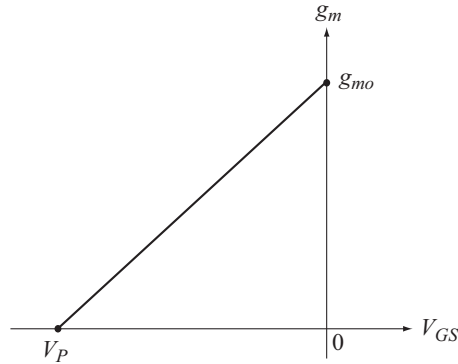


Fig. 9.7 Plot of g_m versus V_{GS}

Example 9.2

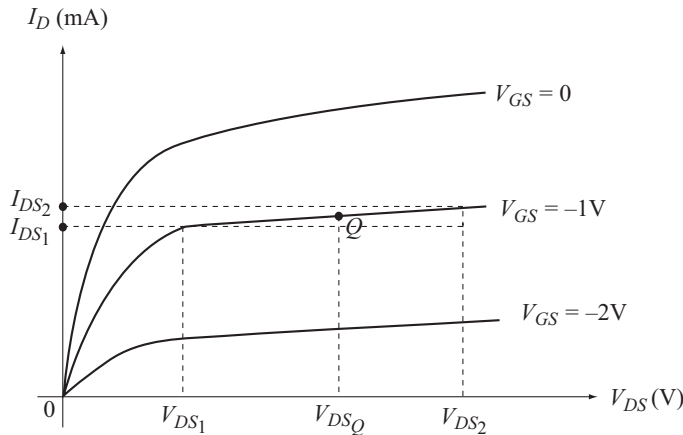
Explain the graphical determination of r_d .

Solution

The output impedance is defined as the slope of the horizontal portion of the drain characteristics at the Q point.

$$r_d = \left. \frac{\Delta V_{DS}}{\Delta I_D} \right|_{V_{GS}=\text{const}}$$

The calculation of r_d is illustrated in the following figure.



$$r_d = \left. \frac{V_{DS_2} - V_{DS_1}}{I_{DS_2} - I_{DS_1}} \right|_{V_{GS}=-1V}$$

Typically r_d is in the range of $20\text{ k}\Omega - 100\text{ k}\Omega$. Under the ideal situation, the curve is perfectly horizontal. As a result, $\Delta I_D = 0$ and hence $r_d = \infty$.

◆ 9.8 RELATION BETWEEN I_D AND g_m

From Shockley's equation we have

$$I_D = I_{DSS} \left[1 - \frac{V_{GS}}{V_p} \right]^2$$

$$\therefore \frac{I_D}{I_{DSS}} = \left[1 - \frac{V_{GS}}{V_p} \right]^2$$

$$\Rightarrow \left[1 - \frac{V_{GS}}{V_p} \right] = \sqrt{\frac{I_D}{I_{DSS}}} \quad (9.16)$$

But

$$g_m = g_{m_0} \left[1 - \frac{V_{GS}}{V_p} \right]$$

$$\therefore g_m = g_{m_0} \sqrt{\frac{I_D}{I_{DSS}}} \quad (9.17)$$

Observe from Equation (9.17) that, g_m varies as the square root of I_D .

◆ 9.9 JFET COMMON-SOURCE AMPLIFIER USING FIXED BIAS CONFIGURATION

JFET common-source amplifier is analogous to BJT common-emitter amplifier. Figure 9.8 shows the circuit of common-source amplifier using fixed bias configuration.

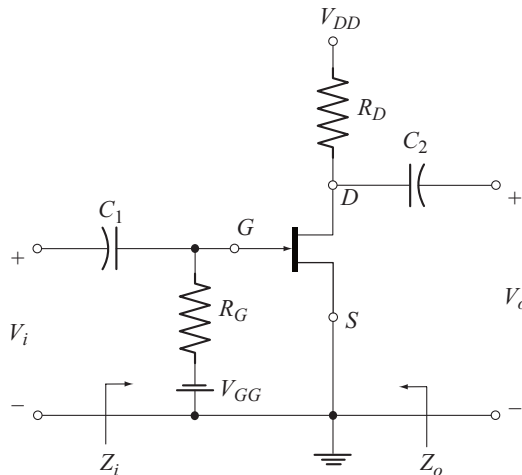


Fig. 9.8 JFET common-source amplifier using fixed bias configuration

Due to high input impedance of JFET, the dc current through R_G is almost zero. Thus a constant bias voltage $-V_{GG}$ is applied between the gate and source terminals. Hence the name fixed bias configuration.

The input coupling capacitor isolates the dc supply $-V_{GG}$ and the ac input signal V_i . The output coupling capacitor isolates the dc supply V_{DD} and the load. They are properly selected so as to represent short-circuits at the lowest frequency of operation.

AC Equivalent Circuit

The ac equivalent circuit is obtained by short circuiting the capacitors C_1 and C_2 and reducing the dc sources V_{DD} and $-V_{GG}$ to zero, as shown in Fig. 9.9.

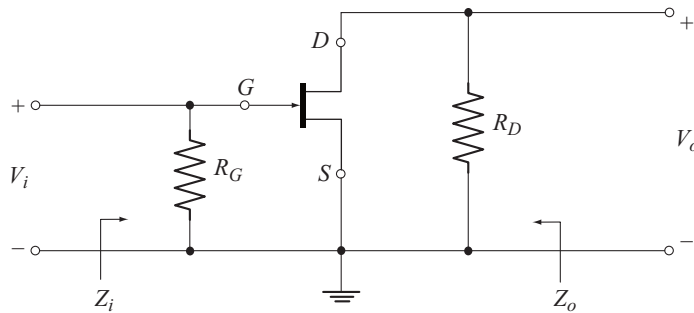


Fig. 9.9 Ac equivalent circuit of JFET common-source amplifier

Let us replace the JFET by its small-signal ac equivalent circuit. The resulting circuit is shown in Fig. 9.10.

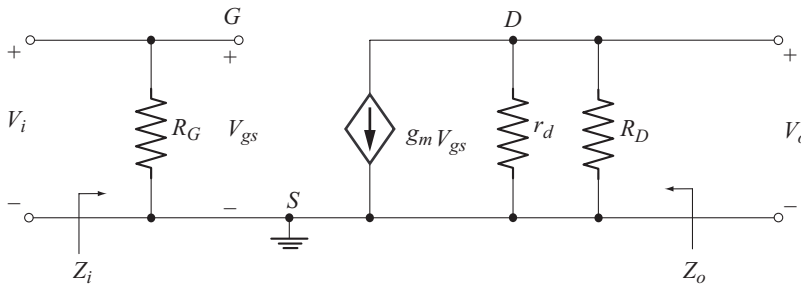


Fig. 9.10 Ac equivalent using small-signal ac model of JFET

Input Impedance (Z_i)

Observe in Fig. 9.10 that, there is an open circuit between the gate and source terminals. Hence the input impedance is nothing but R_G

$$\text{ie} \quad Z_i = R_G \quad (9.18)$$